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1	Extra	
1.1	vimrc	<pre>1 syntax on 2 set ts=4 sw=4 sta et nu rnu sc cindent autoread mouse=a is hls 3 4 inoremap {<CR> {<CR>}<Esc>O 5 nnoremap <C-down> :m+1<CR> 6 nnoremap <C-up> :m-2<CR> 7 8 nnoremap <F5> :w<CR>:!g++ -fsanitize=address -g % -o %: r && ./%:r<CR> 9 nnoremap <F6> :w<CR>:!g++ -fsanitize=address -g % -o %: r && ./%:r < <C-r>=input('file: ')<CR><CR></pre>
2	Data Structures	
2.1	BIT (Fenwick Tree)	Data structure that can handle point update and range queries for invertible operations. Obs: If the operation is not invertible (e.g. min,max,gcd), it can only answer prefix queries <pre>1 // [l,r] interval convention and 0-indexed parameters 2 struct BIT { 3 vector<ll> bit; 4 int n; 5 BIT(int _n = 0){ 6 n = _n; 7 bit.assign(n+1,0ll); 8 } 9 ll presum(int i){ 10 ll ans = 0; 11 for (++i; i > 0; i -= i&-i) 12 ans += bit[i]; 13 return ans; 14 } 15 ll sum(int l, int r){ 16 return presum(r) - presum(l-1); 17 } 18 void add(int i, ll v){ 19 for (++i; i <= n; i += i&-i) 20 bit[i] += v; 21 } 22 int find_kth(int k) { 23 int pos = 0, sum = 0, step = 1; 24 while (step <= n) step <<= 1; 25 step >>= 1; 26 for (; step >= 1; step >>= 1) {</pre>

```

27     if (pos+step <= n && sum + bit[pos+step] <= k) {
28         pos += step;
29         sum += bit[pos];
30     }
31 }
32 return pos;
33 }
34 };

```

2.2 BIT (Fenwick Tree) 2D

2D Sum BIT, update and sum. The struct encapsulates the 1-based indexing, therefore parameters are 0-based indexed.

Query/update time: $\mathcal{O}(\log n \cdot \log m)$

Construction time (using proper constructor): $\mathcal{O}(nm)$

Space: $\mathcal{O}(nm)$

```

1 // [l,r] interval cnvention and 0-based indexing
2 struct BIT2D {
3     vector<ll> bit;
4     int n, m;
5     BIT2D(int _n, int _m){
6         n = _n; m = _m;
7         bit.resize((n+1)*(m+1));
8     }
9     BIT2D(vector<vector<ll>> &a) {
10         n = a.size();
11         m = a[0].size();
12         bit.resize((n+1)*(m+1));
13
14         for (int i = 1; i <= n; i++)
15             for (int j = 1; j <= m; j++)
16                 bit[i*(m+1) + j] = a[i-1][j-1];
17
18         for (int i = 1; i <= n; i++) {
19             for (int j = 1; j <= m; j++) {
20                 int nxt = j + (j&-j);
21                 if (nxt<=m) bit[i*(m+1)+nxt] += bit[i*(m+1)+j];
22             }
23         }
24         for (int i = 1; i <= n; i++) {
25             int nxt = i + (i&-i);
26             if (nxt <= n) {
27                 for (int j = 1; j <= m; j++)
28                     bit[nxt*(m+1) + j] += bit[i*(m+1) + j];
29             }
30         }
31     }
32     void add(int si, int sj, ll v){
33         for(int i = si+1; i <= n; i+=i&-i)
34             for (int j = sj+1; j <= m; j+=j&-j)
35                 bit[i*(m+1) + j] += v;
36     }
37     ll presum(int si, int sj){
38         ll ans = 0;
39         for (int i = si+1; i > 0; i-=i&-i)
40             for (int j = sj+1; j > 0; j-=j&-j)
41                 ans += bit[i*(m+1) + j];
42         return ans;
43     }
44     ll sum(ii ab, ii cd){
45         auto [a,b] = ab;

```

```

46     auto [c,d] = cd;
47     return presum(c,d) - presum(a-1,d) - presum(c,b-1)
48         + presum(a-1,b-1);
49 };

```

2.3 DSU - Disjoint Set Union

Query/update time: $\mathcal{O}(1)$

Construction time: $\mathcal{O}(n)$

Space: $\mathcal{O}(n)$

```

1 struct DSU {
2     vi p, sz;
3     DSU(int n) : p(n), sz(n,1) {
4         iota(p.begin(), p.end(), 0);
5     }
6     int find(int i) {
7         if (p[i] == i) return i;
8         return p[i] = find(p[i]);
9     }
10    bool join(int u, int v) {
11        u = find(u), v = find(v);
12        if (u == v) return false;
13        if (sz[u] < sz[v]) swap(u, v);
14        p[v] = u;
15        sz[u] += sz[v];
16        return true;
17    }
18 };

```

2.4 DSU - Binary Tree

Specific code to find maximum path sums between pairs of vertices. Uses Kruskal-style MST. Query/update time: possibly $\mathcal{O}(n)$ Construction time: $\mathcal{O}(n)$ Space: $\mathcal{O}(n)$

```

1 vi d;
2 vi_ii e;
3 vi ans;
4
5 int merged;
6 vi _p, _leaf, _wei;
7 vvi adj;
8 int _find(int u) { return _p[u] == u ? u : _p[u] = _find(_p[u]); }
9 void _union(int u, int v, int w){
10     u = _find(u);
11     v = _find(v);
12     int merge_ind = merged+n;
13     _p[u] = merge_ind;
14     _p[v] = merge_ind;
15     _leaf[merge_ind] = _leaf[u] + _leaf[v];
16     _wei[merge_ind] = max(_wei[u], _wei[v]);
17     adj[u].push_back(merge_ind);
18     adj[merge_ind].push_back(u);
19     adj[v].push_back(merge_ind);
20     adj[merge_ind].push_back(v);
21     merged++;
22 }
23 void make(){
24     _p = vi(2*n);

```

```

25     for(int i = 0; i < 2*n; i++) _p[i] = i;
26     _leaf = vi(2*n, 1);
27     _wei = vi(2*n);
28     for(int i = 0; i < n; i++) _wei[i] = d[i];
29     merged = 0;
30     adj = vvi(2*n);
31 }
32
33 void dfs(int u, int p){
34     for(auto &v: adj[u]){
35         if(v == p) continue;
36         ans[v] = ans[u] + (_leaf[u] - _leaf[v])*_wei[u];
37         dfs(v, u);
38     }
39 }

```

2.5 Implicit Treap

Randomized BST, handles insertion/deletion, reversals, cyclic shifts and more, all in $\mathcal{O}(\log n)$ per query. Can handle any associative function, just as a segtree can

```

1 mt19937 rng(1337); // change to chrono
2 struct implicit_treap {
3     struct tag {
4         int add = 0;
5         bool has = false;
6     };
7     struct node {
8         node *l = 0, *r = 0;
9         uint32_t y = rng();
10        int sz = 1;
11        bool rev = false;
12        // custom data
13        int val = 0, sum = 0;
14        tag lazy;
15
16        node(int _val) : val(_val), sum(_val) {}
17        void apply(tag t) {
18            if (!t.has) return;
19            val += t.add;
20            sum += t.add * sz;
21            lazy.add += t.add;
22            lazy.has = true;
23        }
24        void push() {
25            if (rev) {
26                swap(l, r);
27                if (l) l->rev ^= 1;
28                if (r) r->rev ^= 1;
29                rev = false;
30            }
31            if (lazy.has) {
32                if (l) l->apply(lazy);
33                if (r) r->apply(lazy);
34                lazy = tag();
35            }
36        }
37        void pull() {
38            sz = 1;
39            sum = val;
40            if (l) { sz += l->sz; sum += l->sum; }
41            if (r) { sz += r->sz; sum += r->sum; }
42        }
43    };
44    node* root = 0;
45    inline int size(node* t) { return t ? t->sz : 0; }

```

```

46 // split node t's first k indices
47 // write output treaps into L and R
48 void split(node* t, int k, node*& L, node*& R) {
49     if (!t) { L = R = 0; return; }
50     t->push();
51     if (size(t->l) >= k) {
52         split(t->l, k, L, t->l);
53         R = t;
54     } else {
55         split(t->r, k - size(t->l) - 1, t->r, R);
56         L = t;
57     }
58     t->pull();
59 }
60 // merge nodes L and R and write it into t
61 void merge(node*& t, node* L, node* R) {
62     if (!L || !R) { t = L ? L : R; return; }
63     L->push(); R->push();
64     if (L->y > R->y) {
65         merge(L->r, L->r, R);
66         t = L;
67     } else {
68         merge(R->l, L, R->l);
69         t = R;
70     }
71     t->pull();
72 }
73 implicit_treap(const vi& a) { // O(N) build
74     vector<node*> st; st.reserve(a.size());
75     for (int x : a) {
76         node *cur = new node(x), *last = 0;
77         while (!st.empty() && st.back()->y < cur->y) {
78             last = st.back(); st.pop_back();
79         }
80         cur->l = last;
81         if (!st.empty()) st.back()->r = cur;
82         st.push_back(cur);
83     }
84     root = st.empty() ? nullptr : st.front();
85     auto dfs = [&](auto &&dfs, node *t) -> void {
86         if (!t) return;
87         dfs(dfs, t->l);
88         dfs(dfs, t->r);
89         t->pull();
90     }; dfs(dfs, root);
91 }
92 void insert(int pos, int val) {
93     node *L, *R;
94     split(root, pos, L, R);
95     merge(L, L, new node(val));
96     merge(root, L, R);
97 }
98 void remove(int pos) {
99     node *L, *mid, *R;
100     split(root, pos, L, R);
101     split(R, 1, mid, R);
102     delete mid;
103     merge(root, L, R);
104 }
105 void rangeUpdate(int l, int r, tag v) {
106     node *L, *mid, *R;
107     split(root, l, L, R);
108     split(R, r - l + 1, mid, R);
109     if (mid) mid->apply(v);
110     merge(R, mid, R);
111     merge(root, L, R);
112 }
113 void reverse(int l, int r) {
114     node *L, *mid, *R;
115     split(root, l, L, R);

```

```

116     split(R, r - l + 1, mid, R);
117     if (mid) mid->rev ^= 1;
118     merge(R, mid, R);
119     merge(root, L, R);
120 }
121 int query(int l, int r) {
122     node *L, *mid, *R;
123     split(root, l, L, R);
124     split(R, r - l + 1, mid, R);
125     int ans = mid ? mid->sum : 0;
126     merge(R, mid, R);
127     merge(root, L, R);
128     return ans;
129 }
130 vi output() { // full array output
131     vi ans; ans.reserve(size(root));
132     auto go = [&](auto &&go, node *t) -> void {
133         if (!t) return;
134         t->push();
135         go(go, t->l);
136         ans.push_back(t->val);
137         go(go, t->r);
138     }; go(go, root);
139     return ans;
140 }
141 void erase_range(int l, int r) {
142     node *L, *mid, *R;
143     split(root, l, L, R);
144     split(R, r - l + 1, mid, R);
145     // care for mem leaks, destroy mid recursively if
146     // needed
147     merge(root, L, R);
148 }
149 void insert_range(int pos, node* new_treap) {
150     node *L, *R;
151     split(root, pos, L, R);
152     merge(L, L, new_treap);
153     merge(root, L, R);
154 };

```

2.6 MinQueue

Minimum (or maximum) queue. All operations are $\mathcal{O}(1)$ on average. Useful for sliding window max/min queries

```

1 template<typename T>
2 struct MinQueue {
3     deque<pair<T, int>> q;
4     int added = 0;
5     int removed = 0;
6     // returns [value, index]
7     pair<T, int> getmin() { return q.front(); }
8
9     void push(T x) {
10         // < for MaxQueue
11         while (!q.empty() && q.back().first > x) q.
12             pop_back();
13         q.push_back({x, added++});
14     }
15     void pop() {
16         if (!q.empty() && q.front().second == removed) q.
17             pop_front();
18         removed++;
19     }
20     bool empty() { return added == removed; }
21 };

```

2.7 Mo's Algorithm

A technique for solving offline range queries on static arrays by sorting queries to minimize total pointer movement. It processes intervals by incrementally updating the range via `add` and `remove` operations. With the optimal block size, the time complexity is $\mathcal{O}((N + Q)\sqrt{N})$ or $\mathcal{O}(N\sqrt{Q})$, depending on block size choice ($\sqrt{N}$ or $N/\sqrt{Q}$). This example solves queries for distinct elements in range

```

1 struct Mo {
2     struct Query {
3         int l, r, idx, b;
4         bool operator<(const Query& o) const {
5             return b != o.b ? b < o.b :
6                 (b & 1 ? r > o.r : r < o.r);
7         }
8     };
9     int n, block_sz;
10    vi freq, a; // custom stuff
11    int ans = 0;
12    vector<Query> queries;
13    Mo(int n) : n(n), block_sz(round(sqrt(n))) {}
14    // [l,r] indexed
15    void add_query(int l, int r, int i) {
16        queries.push_back({l, r, i, l/block_sz});
17    }
18    void add(int i) { // add val at i
19        freq[a[i]]++;
20        if (freq[a[i]] == 1) ans++;
21    }
22    void remove(int i) { // remove value at i
23        freq[a[i]]--;
24        if (freq[a[i]] == 0) ans--;
25    }
26    int get_ans() { // compute current answer
27        return ans;
28    }
29    vi run() {
30        vi ans(queries.size());
31        sort(queries.begin(), queries.end());
32        int l = 0, r = -1;
33        for (auto& q : queries) {
34            while (l > q.l) add(--l);
35            while (r < q.r) add(++r);
36            while (l < q.l) remove(l++);
37            while (r > q.r) remove(r--);
38            ans[q.idx] = get_ans();
39        }
40        return ans;
41    }
42 };

```

2.8 Mo with updates

Also sorts queries to minimize total pointer movement, but now there is an added dimension for time (POINT updates). $\mathcal{O}(QN^{2/3})$

```

1 struct MoUpdate {
2     int n;
3     const int block_sz;
4     struct Query {
5         int l, r, t, id;

```

```

6   };
7   vector<Query> queries;
8   vector<ii> updates;
9   vi a;
10
11   MoUpdate(vector<int> &v, int q) : block_sz(round(cbrt(
12       (2*v.size()*v.size()))) {
13       n = v.size() + q;
14       a = v;
15       init();
16   }
17   void add_query(int l, int r) {
18       queries.push_back({l,r,(int)updates.size(),(int)
19           queries.size()});
20   }
21   void add_update(int i, int x) {
22       updates.push_back({i,x});
23   }
24   void update(int l, int r, int t) {
25       auto &[i,v] = updates[t];
26       if (l <= i && i <= r) remove(i);
27       swap(a[i],v);
28       if (l <= i && i <= r) add(i);
29   }
30   vector<int> run() {
31       vector<int> ans(queries.size());
32       sort(queries.begin(),queries.end(), [&](Query qa,
33           Query qb){
34           int l = qa.l/block_sz;
35           int r = qa.r/block_sz;
36           int ql = qb.l/block_sz;
37           int qr = qb.r/block_sz;
38           using iii = tuple<int,int,int>;
39           return iii(l, l&1 ? -r:r, r&1 ? qa.t:-qa.t) <
40               iii(ql, ql&1 ? -qr:qr, qr&1 ? qb.t:-qb.t);
41       });
42       int l = 0, r = -1, t = 0;
43       for (auto &q : queries) {
44           while(t < q.t) update(l,r,t++);
45           while(t > q.t) update(l,r,--t);
46           while(l > q.l) add(--l);
47           while(r < q.r) add(++r);
48           while(l < q.l) remove(l++);
49           while(r > q.r) remove(r--);
50           ans[q.id] = get_ans();
51       }
52       return ans;
53   }
54   int ans = 0; // custom stuff
55   void init(){} // initialize custom stuff
56   void remove(int i) {} // remove a[i] from ans
57   void add(int i) {} // add a[i] to ans
58   int get_ans() {}
59 };
```

2.9 Policy Based Data Structures

Ordered Set and ordered multiset, possible to insert/erase and query index. All in $\mathcal{O}(\log n)$.

```

1 #include <ext/pb_ds/assoc_container.hpp>
2 #include <ext/pb_ds/tree_policy.hpp>
3 using namespace __gnu_pbds;
4 using ordered_set = tree<int,null_type,less<int>,
5     rb_tree_tag,
6     tree_order_statistics_node_update>;
7 // multiset: store {value, unique_id}
```

```

7 using ordered_multiset = tree<pair<ii,null_type,less<ii>,
8     rb_tree_tag,tree_order_statistics_node_update>;
9
10 // set:
11 s.insert(x); s.erase(x);
12 s.order_of_key(x); // # < x
13 *s.find_by_order(k); // k-th smallest (0-indexed)
14
15 // multiset:
16 ms.order_of_key({x,0}); // # < x
17 ms.order_of_key({x,INF}); // # <= x
18 ms.find_by_order(k)->first; // k-th value
```

2.10 Segment Tree

You should (almost) never alter the `segtree` logic. Ideally you only modify the `node` struct by answering four questions:

1. **tag**: What data does a range update carry? (e.g., a single integer for addition, or a pair of integers a, d for an AP).
2. **apply()**: How does a **tag** mathematically alter the state of a single node?
3. **push()**: How does a parent node transfer its pending **tag** to its left and right children?
4. **combine()**: How do you calculate the state of a parent by merging the states of its children?

OBS: lazy segtrees must be called with `has=true`

2.10.1 Iterative Point Update

Best for: Standard point updates and range queries (e.g., Point Add + Range Sum).

- **Performance:** Best possible.
- **Memory:** Best possible: $2n$ allocation.
- **Use Case:** If no range updates, then use this.

```

1 // ITERATIVE - implemented for sum + point assignment
2 struct segtree {
3     struct node {
4         int sum = 0; // data
5         static node combine(const node& a, const node& b) {
6             return {a.sum + b.sum};
7         }
8         void apply(int v) {
9             sum = v;
10        }
11    };
12    int n;
13    vector<node> t;
14    void init(const vi& a) {
15        n = a.size();
16        t.assign(2*n, node());
17        for (int i = 0; i < n; i++) t[n + i].apply(a[i]);
```

```

18         for (int i = n-1; i > 0; i--) t[i] = node::combine(
19             t[i<<1], t[i<<1|1]);
20     }
21     void set(int p, int v) {
22         for (t[p += n].apply(v), p >= 1; p > 0; p >= 1) {
23             t[p] = node::combine(t[p<<1], t[p<<1|1]);
24         }
25     }
26     node query(int l, int r) {
27         node resL, resR;
28         for (l+=n, r+=n; l < r; l>=1, r>=1) {
29             if (l&1) resL = node::combine(resL, t[l++]);
30             if (r&1) resR = node::combine(t[--r], resR);
31         }
32         return node::combine(resL, resR);
33     };
```

2.10.2 Iterative Lazy

Best for: Index-independent range updates (e.g., Range Add, Range Set, Range Max, Range Flip).

- **Performance:** Noticeably faster than recursive lazy trees.
- **Index Blindness:** If the update depends on the range indexes themselves (such as AP addition), use the recursive lazy one.

```

1 // LAZY ITERATIVE - implemented for Hash + range
2 // assignment
3 // for reference, see DoubleHashing implementation
4 // always call rangeUpdate with true on tag
5 struct segtree {
6     struct node {
7         struct tag {
8             int val = 0; // make sure default values are
9                 correct
10            bool has = false;
11        };
12        Hash hash; // custom stuff
13        int len = 0;
14        tag lazy;
15        // combine custom stuff
16        static node combine(const node &a, const node &b){}
17        void apply(tag t){
18            if (!t.has) return;
19            // apply tag to custom stuff
20            lazy = t; // remember to upd lazy tag
21        }
22        void push(node &left, node &right){
23            if (!lazy.has) return;
24            left.apply(lazy);
25            right.apply(lazy);
26            lazy = tag();
27        }
28        void pull(const node &left, const node &right){
29            tag cur = lazy;
30            *this = combine(left,right);
31            apply(cur);
32        }
33    };
34    int n, h;
35    vector<node> t;
36    segtree(const string& a) {
```

```

35 int sz = a.size();
36 n = 1; h = 0;
37 while (n < sz) { n*=2; h++; }
38
39 t.assign(2*n, node());
40
41 for (int i = 0; i < sz; i++) {
42     t[i+n].len = 1; // init custom stuff
43     t[i+n].hash.len = 1;
44     t[i+n].apply({a[i], true});
45 }
46 for (int i = n-1; i > 0; i--) t[i].pull(t[i<<1], t[
47     i<<1|1]);
48
49 void build(int p) {
50     while (p > 1) p >>= 1, t[p].pull(t[p<<1], t[p
51         <<1|1]);
52 }
53 void push(int p) {
54     for (int s = h; s > 0; s--) {
55         int i = p >> s;
56         t[i].push(t[i<<1], t[i<<1|1]);
57     }
58 }
59 void rangeUpdate(int l, int r, node::tag v) {
60     if (l==r) return;
61     l += n, r += n;
62     int l0 = l, r0 = r;
63
64     push(l0); push(r0-1);
65     for (; l < r; l >>= 1, r >>= 1) {
66         if (l&1) t[l++].apply(v);
67         if (r&1) t[--r].apply(v);
68     }
69     build(l0); build(r0-1);
70 }
71 node query(int l, int r) {
72     if (l == r) return node();
73     l += n, r += n;
74     push(l); push(r-1);
75     node resL = node(), resR = node();
76     for (; l < r; l >>= 1, r >>= 1) {
77         if (l&1) resL = node::combine(resL, t[l++]);
78         if (r&1) resR = node::combine(t[--r], resR);
79     }
80     return node::combine(resL, resR);
81 };

```

2.10.3 Persistent

```

1 // PERSISTENT - implemented for sum + point assignment
2 struct persistent_segtree {
3     struct node {
4         int sum = 0;
5         int lc = 0, rc = 0; // indices to left and right
6         children
7         static node combine(const node& left, const node&
8             right) {
9             return {left.sum+right.sum};
10        }
11        void leaf(int v) { sum = v; }
12    };
13    int size;
14    vector<node> t;
15    vi root; // root node id for each version
16    persistent_segtree(const vi &a) {
17        size = 1;

```

```

16 while (size < a.size()) size <= 1;
17 t.push_back(node());
18 root.push_back(build(0, size, a));
19 }
20 int build(int lx, int rx, const vi &a) {
21     int x = t.size();
22     t.push_back(node());
23     if (rx-lx == 1) {
24         if (lx < a.size()) t[x].leaf(a[lx]);
25         return x;
26     }
27     int mx = (lx+rx)/2;
28     t[x].lc = build(lx, mx, a);
29     t[x].rc = build(mx, rx, a);
30     // do not override pointers lc,rc
31     t[x].sum = node::combine(t[t[x].lc], t[t[x].rc]).
32         sum;
33     return x;
34 }
35 int update(int i, int v, int old, int lx, int rx) {
36     int x = t.size();
37     t.push_back(t[old]);
38     if (rx-lx == 1) {
39         t[x].leaf(v);
40         return x;
41     }
42     int mx = (lx+rx)/2;
43     if (i < mx) t[x].lc = update(i, v, t[old].lc, lx,
44         mx);
45     else t[x].rc = update(i, v, t[old].rc, mx, rx);
46     // do not override pointers lc,rc
47     t[x].sum = node::combine(t[t[x].lc], t[t[x].rc]).
48         sum;
49     return x;
50 }
51 int update(int version, int i, int v) {
52     // assign to root[version] if this update
53     // should not trigger another version
54     int x = update(i, v, root[version], 0, size);
55     root.push_back(x);
56     return root.size()-1;
57 }
58 int copy_version(int version) {
59     root.push_back(root[version]);
60     return root.size() - 1;
61 }
62 node query(int l, int r, int x, int lx, int rx) {
63     if (lx >= r || rx <= l || x == 0) return node();
64     if (lx >= l && rx <= r) return t[x];
65     int mx = (lx+rx)/2;
66     return node::combine(
67         query(l, r, t[x].lc, lx, mx),
68         query(l, r, t[x].rc, mx, rx)
69     );
70 }
71 node query(int version, int l, int r) {
72     return query(l, r, root[version], 0, size);
73 };

```

2.10.4 Recursive Lazy

Best for: Index-dependent updates (e.g. AP addition), Tree Walking (Binary Search on Tree), and persistency.

- **Performance:** It is the slowest of the three, use only if necessary.

```

1 // LAZY recursive - implemented for sum + range AP
2 addition
3 struct segtree {
4     struct node {
5         struct tag {
6             int a = 0, d = 0; // be sure default is correct
7             bool has = false;
8         };
9         int sum = 0; // data
10        tag lazy;
11        static node combine(const node& a, const node& b) {
12            return {a.sum + b.sum, {0, 0, false}};
13        }
14        void leaf(int v) { sum = v; }
15        void apply(tag t, int lx, int rx) {
16            if (!t.has) return;
17            int len = rx - lx;
18
19            // Sum of (t.a + t.d * i) for i in [lx, rx-1]
20            sum += t.a * len + t.d*len*(lx+rx-1)/2;
21
22            lazy.a += t.a;
23            lazy.d += t.d;
24            lazy.has = true;
25        }
26        void push(node& left, node& right, int lx, int rx)
27        {
28            if (lazy.has) {
29                int mx = (lx + rx) / 2;
30                left.apply(lazy, lx, mx);
31                right.apply(lazy, mx, rx);
32                lazy = tag();
33            }
34        };
35        int size;
36        vector<node> t;
37        void init(int n) {
38            size = 1;
39            while (size < n) size *= 2;
40            t.assign(2*size, node());
41            build(0, 0, size, n);
42        }
43        void build(int x, int lx, int rx, int n) {
44            if (rx-lx == 1) {
45                if (lx < n) t[x].leaf(0);
46                return;
47            }
48            int mx = (lx + rx) / 2;
49            build(2*x+1, lx, mx, n);
50            build(2*x+2, mx, rx, n);
51            t[x] = node::combine(t[2*x+1], t[2*x+2]);
52        }
53        void push(int x, int lx, int rx) {
54            t[x].push(t[2*x+1], t[2*x+2], lx, rx);
55        }
56        void rangeUpdate(int l, int r, node::tag v, int x,
57            int lx, int rx) {
58            if (lx >= r || rx <= l) return;
59            if (lx >= l && rx <= r) {
60                t[x].apply(v, lx, rx);
61                return;
62            }
63            push(x, lx, rx);
64            int mx = (lx + rx) / 2;
65            rangeUpdate(l, r, v, 2*x+1, lx, mx);
66            rangeUpdate(l, r, v, 2*x+2, mx, rx);
67            t[x] = node::combine(t[2*x+1], t[2*x+2]);
68        }
69        void rangeUpdate(int l, int r, int a, int d) {

```

```

68     rangeUpdate(l, r, {a - l*d, d, true}, 0, 0, size);
69 }
70 node query(int l, int r, int x, int lx, int rx) {
71     if (lx >= r || rx <= l) return node();
72     if (lx >= l && rx <= r) return t[x];
73     push(x, lx, rx);
74     int mx = (lx+rx) / 2;
75     return node::combine(
76         query(l, r, 2*x+1, lx, mx),
77         query(l, r, 2*x+2, mx, rx)
78     );
79 }
80 node query(int l, int r) {
81     return query(l, r, 0, 0, size);
82 }
83 };

```

2.11 Sparse Table RMQ

Sparse table for RMQ in $\mathcal{O}(1)$, used in many problems, including $\mathcal{O}(1)$ LCA (Trees) and LCP (SuffixArray) queries.

```

1 struct SparseTable {
2     vvi st;
3     SparseTable(const vi &a) {
4         int n = a.size();
5         int max_log = __lg(n)+1;
6         st.assign(max_log, vector<i>(n));
7         for (int i = 0; i < n; i++) {
8             st[0][i] = {a[i], i};
9         }
10        for (int i = 1; i < max_log; i++) {
11            for (int j = 0; j + (1 << i) <= n; j++) {
12                // Combine the two halves
13                st[i][j] = f(st[i-1][j], st[i-1][j+(1<<(i-1))]);
14            }
15        }
16    }
17    // Returns (min, id) in range [l, r] inclusive
18    ii min(int l, int r) {
19        int len = r - l + 1;
20        int k = __lg(len);
21        return f(st[k][l], st[k][r - (1<<k) + 1]);
22    }
23    inline ii f(ii l, ii r) { return std::min(l, r); }
24 };

```

3 Graphs

3.1 Block Cut Tree (BCT)

A data structure that condenses an undirected graph into a bipartite graph using two types of nodes: B-nodes (biconnected components) and C-nodes (articulation points). Edges strictly connect a B-node to a C-node if the articulation point belongs to that component. Yields exactly one tree per connected component.

Articulation points are explicitly represented as the C-nodes. Because an articulation point is a vertex that dis-

connects a component upon removal, it acts as a literal bridge in the BCT. Any path navigating between two distinct biconnected components in the original graph is forced to route through the shared C-node.

```

1 vvi build_bct(const vvi &adj){
2     int n = adj.size();
3     vi in(n, -1), low(n), st;
4     int timer = 0;
5     vvi bccs;
6
7     auto dfs = [&](auto &&dfs, int u, int p) -> void {
8         in[u] = low[u] = ++timer;
9         st.push_back(u);
10        if (p == -1 && adj[u].empty()){
11            bccs.push_back({u});
12            st.pop_back();
13            return;
14        }
15        for (int v : adj[u]) if (v != p) {
16            if (in[v] != -1) {
17                low[u] = min(low[u], in[v]);
18            } else {
19                dfs(dfs, v, u);
20                low[u] = min(low[u], low[v]);
21                if (low[v] >= in[u]){
22                    bccs.push_back({u});
23                    while(1){
24                        int w = st.back();
25                        st.pop_back();
26                        bccs.back().push_back(w);
27                        if (w == v) break;
28                    }
29                }
30            }
31        }
32    };
33    for (int i = 0; i < n; i++){
34        if (in[i] == -1) dfs(dfs, i, -1);
35    }
36    vvi bct(n+bccs.size());
37    for (int i = 0; i < bccs.size(); i++){
38        int b = n+i;
39        for (int u : bccs[i]){
40            bct[u].push_back(b);
41            bct[b].push_back(u);
42        }
43    }
44    return bct;
45 }

```

3.2 Bridges and Articulation points

DFS to get bridges and articulation points of a graph in $\mathcal{O}(n + m)$

```

1 vvi adj;
2 vi in, low;
3 int timer = 1; // start in 1
4 set<int> aps;
5 vector<ii> bridges;
6
7 void dfs_ap(int u, int p = -1) {
8     in[u] = low[u] = ++timer;
9     int ch = 0;
10    for (int v : adj[u]) {

```

```

11        if (v == p) continue;
12        if (in[v]) {
13            // Back-edge
14            low[u] = min(low[u], in[v]);
15        } else {
16            // Tree-edge
17            dfs_ap(v, u);
18            low[u] = min(low[u], low[v]);
19            if (low[v] >= in[u] && p != -1)
20                aps.insert(u);
21            ch++;
22        }
23    }
24    if (p == -1 && ch > 1)
25        aps.insert(u);
26 }
27
28 void dfs_bridges(int u, int p = -1) {
29     in[u] = low[u] = ++timer;
30     for (int v : adj[u]) {
31         if (v == p) continue;
32         if (in[v]) {
33             low[u] = min(low[u], in[v]);
34         } else {
35             dfs_bridges(v, u);
36             low[u] = min(low[u], low[v]);
37             if (low[v] > in[u])
38                 bridges.push_back({u, v});
39         }
40     }
41 }

```

3.3 Dinic - Flow/matchings

- **General Network:** $\mathcal{O}(VE \log U)$.
- **Unit Capacity Network:** $\mathcal{O}(\min(V^{2/3}, E^{1/2}) \cdot E)$. Often considered $\mathcal{O}(E\sqrt{V})$.
- **Bipartite Matching:** $\mathcal{O}(\min(V^{2/3}, E^{1/2}) \cdot E)$. Often considered $\mathcal{O}(E\sqrt{V})$.

```

1 struct Dinic {
2     struct Edge {
3         int u, v;
4         ll cap, flow = 0;
5         Edge(int u, int v, ll cap) : u(u), v(v), cap(cap) {}
6     };
7     const ll flow_inf = 1e18;
8     vector<Edge> edges;
9     vvi adj;
10    int n, m = 0;
11    int s, t;
12    vi level, ptr;
13    queue<int> q;
14    Dinic(int n): n(n) {
15        adj.resize(n);
16        level.resize(n);
17        ptr.resize(n);
18    }
19    void add_edge(int u, int v, ll cap) {
20        edges.emplace_back(u, v, cap);
21        edges.emplace_back(v, u, 0);
22        adj[u].push_back(m++);

```

```

23     adj[v].push_back(m++);
24 }
25 bool bfs(ll delta){
26     queue<int> q;
27     q.push(s);
28     while(!q.empty()){
29         int u = q.front(); q.pop();
30         for (int id : adj[u]){
31             auto &e = edges[id];
32             if (e.cap - e.flow < delta) continue;
33             if (level[e.v] != -1) continue;
34             level[e.v] = level[u]+1;
35             q.push(e.v);
36         }
37     }
38     return level[t] != -1;
39 }
40 ll dfs(int u, ll pushed) {
41     if (pushed == 0) return 0;
42     if (u == t) return pushed;
43     for (int &cid = ptr[u]; cid < (int)adj[u].size();
44         cid++){
45         int id = adj[u][cid];
46         auto &e = edges[id];
47         if (level[u]+1 != level[e.v]) continue;
48         ll tr = dfs(e.v, min(pushed, e.cap - e.flow));
49         if (tr == 0) continue;
50         e.flow += tr;
51         edges[id^1].flow -= tr;
52         return tr;
53     }
54     return 0;
55 }
56 ll maxflow(int s, int t){
57     this->s = s; this->t = t;
58     ll max_c = 0;
59     for (auto &e : edges) max_c = max(max_c, e.cap);
60     ll delta = 1;
61     while(delta <= max_c) delta <= 1;
62     delta >= 1;
63     ll f = 0;
64     for (; delta > 0; delta >= 1){
65         while(true){
66             fill(level.begin(), level.end(), -1);
67             level[s] = 0;
68             if (!bfs(delta)) break;
69             fill(ptr.begin(), ptr.end(), 0);
70             while(ll pushed = dfs(s, flow_inf)) f += pushed;
71         }
72     }
73     return f;
74 }
75 // call constructor with (n1+n2+2) beforehand (dont
76 // add edges manually)
77 // assumes pairs are 1-indexed
78 vii maxmatchings(int n1, int n2, const vii& pairs){
79     for (int i = 1; i <= n1; i++)
80         add_edge(0, i, 1);
81     for (int i = 1; i <= n2; i++)
82         add_edge(i+n1, n-1, 1);
83     for (auto &[u, v] : pairs)
84         add_edge(u, v+n1, 1);
85     maxflow(0, n-1);
86     vii matchings;
87     for (auto &e : edges){
88         if (e.u >= 1 && e.u <= n1 && e.flow == 1 && e.v >
89             n1){
90             matchings.emplace_back(e.u, e.v-n1);
91         }
92     }
93 }

```

```

90     return matchings;
91 }
92 vii mincut(int s, int t){
93     maxflow(s, t);
94     queue<int> q; q.push(s);
95     vector<bool> reachable(n);
96     reachable[s] = true;
97     while(!q.empty()){
98         int u = q.front(); q.pop();
99         for (auto &id : adj[u]){
100             int v = edges[id].v;
101             if (edges[id].cap - edges[id].flow > 0 && !
102                 reachable[v]) {
103                 reachable[v] = true;
104                 q.push(v);
105             }
106         }
107     }
108     vii minCutEdges;
109     for (int i = 0; i < m; i += 2) {
110         const Edge& edge = edges[i];
111         if (reachable[edge.u] && !reachable[edge.v]) {
112             minCutEdges.emplace_back(edge.u, edge.v);
113         }
114     }
115     return minCutEdges;
116 }

```

3.4 Euler path/tour - directed/undirected

3.4.1 Euler path/tour - directed

```

1 auto get_path = [&](int s = -1, int t = -1) -> vi {
2     if (m == 0) return {};
3     if (s != -1 && t != -1){
4         if (out[s]-in[s] != 1 || out[t]-in[t] != -1) return
5         {};
6     }
7     int cnt1 = 0, cnt2 = 0;
8     for (int u = 0; u < n; u++) {
9         int d = out[u]-in[u];
10        if (d == 1) {
11            if (s == -1) s = u;
12            cnt1++;
13        } else if (d == -1) {
14            cnt2++;
15            if (t == -1) t = u;
16        } else if (in[u] != out[u]) return {};
17    }
18    if (!(cnt1==0 && cnt2==0) || (cnt1==1 && cnt2==1))
19        return {};
20    if (s == -1) {
21        for (int i = 0; i < n; i++) {
22            if (out[i] > 0) {
23                s = i; break;
24            }
25        }
26    }
27    vi path;
28    auto dfs = [&](auto& dfs, int u) -> void {
29        while (!adj[u].empty()) {
30            int v = adj[u].back();
31            adj[u].pop_back();
32            dfs(dfs, v);
33        }
34        path.push_back(u);
35    }

```

```

35    }; dfs(dfs, s);
36    if (path.size() != m+1) return {};
37    reverse(all(path));
38    return path;
39 };
40 vi path = get_path(0, n-1);
41 if (path.empty()) {
42     cout << "IMPOSSIBLE\n"; return;
43 }
44 for (int u : path) cout << u+1 << ' '; cout << '\n';

```

3.4.2 Euler path/tour - undirected

```

1 auto get_path = [&](int s = -1, int t = -1) -> vi {
2     if (m == 0) return {};
3     if (s != -1 && t != -1) {
4         if (s != t && (deg[s]%2 == 0 || deg[t]%2 == 0))
5             return {};
6         if (s == t && deg[s]%2) return {};
7     }
8     int odd = 0;
9     for (int u = 0; u < n; u++) {
10        if (deg[u]&1) {
11            odd++;
12            if (s == -1) s = u;
13            else if (t == -1 && s != u) t = u;
14        }
15    }
16    if (odd != 0 && odd != 2) return {};
17    if (s == -1) {
18        for (int u = 0; u < n; u++) {
19            if (deg[u] > 0) {
20                s = u; break;
21            }
22        }
23    }
24    if (odd == 2 && deg[s]%2 == 0) return {};
25    if (odd == 2 && deg[t]%2 == 0) return {};
26    vi path;
27    vector<bool> used(m, false);
28    auto dfs = [&](auto& dfs, int u) -> void {
29        while (!adj[u].empty()) {
30            auto [v, id] = adj[u].back();
31            adj[u].pop_back();
32            if (used[id]) continue;
33            used[id] = true;
34            dfs(dfs, v);
35        }
36        path.push_back(u);
37    }; dfs(dfs, s);
38    if (path.size() != m+1) return {};
39    reverse(all(path));
40    return path;
41 };
42 vi path = get_path(0, 0);
43 if (path.empty()) {
44     cout << "IMPOSSIBLE\n"; return;
45 }

```

3.5 Floyd-Warshall

Time: $\mathcal{O}(n^3)$

```

1
2 vvi d(n, vi(n, INF));
3 void floyd_warshall(){
4     for (int k = 0; k < n; k++)
5         for (int i = 0; i < n; i++)

```

```

6     for (int j = 0; j < n; j++)
7         d[i][j] = min(d[i][j], d[i][k]+d[k][j]);
8 }

```

3.6 Hopcroft-Karp - Bipartite Matching

Bipartite matching such as Kuhn but faster. BFS until first layer missing match, DFS for the BFS graph to find pairings. Time: $\mathcal{O}(E\sqrt{V})$

```

1 int n, m, k;
2 vvi adj;
3 vi p, dist; /*p is in matching for [0, n[ and parent
   for [n, n+m[*/
4
5 int bfs(){
6     queue<int> q;
7     dist = vi(n+m, inf);
8     for(int i = 0; i < n; i++){
9         if(p[i] == -1) q.push(i), dist[i] = 0;
10    }
11    int min_dist_match = inf;
12    while(!q.empty()){
13        int u = q.front(); q.pop();
14        if(dist[u] > min_dist_match) continue;
15        for(auto v: adj[u]){
16            if(p[v] == -1) min_dist_match = dist[u];
17            else if(dist[p[v]] == inf){
18                dist[p[v]] = dist[u] + 1;
19                q.push(p[v]);
20            }
21        }
22    }
23    return min_dist_match != inf;
24 }
25
26 int dfs(int u){
27     for(auto v: adj[u]){
28         if(p[v] == -1 || (dist[u]+1 == dist[p[v]] && dfs(p[v])))
29             p[v] = u;
30         p[u] = 1;
31         return true;
32     }
33     dist[u] = inf;
34     return false;
35 }
36
37 int hopkarp(){
38     p = vi(n+m, -1);
39     int matchings = 0;
40     while(bfs()){
41         for(int i = 0; i < n; i++){
42             if(p[i] == -1 && dfs(i)) matchings++;
43         }
44     }
45     return matchings;
46 }
47
48 void create(){
49     adj = vvi(n+m);
50     for(int i = 0; i < k; i++){
51         int u, v;
52         cin >> u >> v; u--; v--;
53         v += n;
54         adj[u].push_back(v);
55     }

```

3.7 Hungarian

Solves minimum cost assignment for n workers and m jobs. Time: $\mathcal{O}((n+m)^3)$

```

1 // cost should be (cost[worker][job])
2 // hungarian(int n, int m, const vvi &cost)
3 {
4     if (n == 0) return {0, {}};
5     int N = max(n, m);
6
7     vi u(N+1), v(N+1), p(N+1), way(N+1);
8
9     const int INF = 1e9;
10    for (int i = 1; i <= n; ++i) {
11        p[0] = i;
12        int j0 = 0;
13        vi minv(N + 1, INF);
14        vector<bool> used(N + 1, false);
15
16        do {
17            used[j0] = true;
18            int i0 = p[j0], delta = INF, j1;
19
20            for (int j = 1; j <= N; ++j) {
21                if (!used[j]) {
22                    int cur = cost[i0-1][j-1] - u[i0] - v[j];
23                    if (cur < minv[j]) {
24                        minv[j] = cur;
25                        way[j] = j0;
26                    }
27                    if (minv[j] < delta) {
28                        delta = minv[j];
29                        j1 = j;
30                    }
31                }
32            }
33
34            for (int j = 0; j <= N; ++j) {
35                if (used[j]) {
36                    u[p[j]] += delta;
37                    v[j] -= delta;
38                } else {
39                    minv[j] -= delta;
40                }
41            }
42            j0 = j1;
43        } while (p[j0] != 0);
44
45        do {
46            int j1 = way[j0];
47            p[j0] = p[j1];
48            j0 = j1;
49        } while (j0);
50    }
51
52    int total_cost = 0;
53    for (int j = 1; j <= m; ++j) {
54        if (p[j] != 0) {
55            total_cost += cost[p[j] - 1][j - 1];
56        }
57    }
58 }

```

```

59 // {worker, job}[] 0-indexed
60 vvi matchings;
61 for (int j = 1; j <= m; ++j) {
62     if (p[j] != 0) {
63         matchings.push_back({p[j] - 1, j - 1});
64     }
65 }
66 return {total_cost, matchings};
67 }

```

3.8 Min cost flow

Time: $\mathcal{O}(FE \log V)$

If negative costs are needed (maximize cost), need to run SPFA once at the start, making the solution $\mathcal{O}(EV + FE \log V)$.

```

1 struct MinCostFlow {
2     struct Edge {
3         int to, cap, rev;
4         ll cost;
5     };
6     int n;
7     vector<vector<Edge>> adj;
8     MinCostFlow(int _n) : n(_n), adj(_n) {}
9     void add_edge(int u, int v, int cap, ll cost){
10         adj[u].push_back({v, cap, sz(adj[v]), cost});
11         adj[v].push_back({u, 0, sz(adj[u]) - 1, -cost});
12     }
13     // O(FE log(V))
14     lli min_cost_flow(int s, int t, int targetFlow) {
15         int flow = 0;
16         ll total_cost = 0;
17         vll dist, h(n);
18         vi pv, pe;
19         spfa(s, h, pv, pe); // only if < 0 costs exist
20         while (flow < targetFlow) {
21             djks(s, h, dist, pv, pe);
22             if (dist[t] == INF) break;
23             for (int i = 0; i < n; i++) {
24                 if (dist[i] < INF) {
25                     h[i] += dist[i];
26                 }
27             }
28             int f = targetFlow - flow;
29             int cur = t;
30             while (cur != s) {
31                 f = min(f, adj[pv[cur]][pe[cur]].cap);
32                 cur = pv[cur];
33             }
34             flow += f;
35             total_cost += f * h[t];
36             cur = t;
37             while (cur != s) {
38                 Edge &e = adj[pv[cur]][pe[cur]];
39                 e.cap -= f;
40                 adj[e.to][e.rev].cap += f;
41                 cur = pv[cur];
42             }
43         }
44         return {total_cost, flow};
45     }
46     // needed only if negative costs exists
47     void spfa(int s, vll &dist, vi &pv, vi &pe) {
48         dist.assign(n, INF);
49         pv = pe = vi(n, -1);

```

```

50 vector<bool> inq(n, false);
51 queue<int> q; q.push(s);
52 dist[s] = 0; inq[s] = true;
53 while (!q.empty()) {
54     int u = q.front(); q.pop();
55     inq[u] = false;
56     for (int i = 0; i < adj[u].size(); i++) {
57         Edge &e = adj[u][i];
58         int v = e.to;
59         if (e.cap > 0 && dist[v] > dist[u] + e.cost) {
60             dist[v] = dist[u] + e.cost;
61             pv[v] = u;
62             pe[v] = i;
63             if (!inq[v]) {
64                 inq[v] = true;
65                 q.push(v);
66             }
67         }
68     }
69 }
70 }
71 void djc(int s, vll &h, vll &dist, vi &pv, vi &pe) {
72     dist.assign(n, INF);
73     pv = pe = vi(n, -1);
74     dist[s] = 0;
75     priority_queue<lli, vector<lli>, greater<lli>> pq;
76     pq.emplace(0, s);
77     while (!pq.empty()) {
78         auto [d, u] = pq.top(); pq.pop();
79         if (d > dist[u]) continue;
80         for (int i = 0; i < adj[u].size(); i++) {
81             Edge &e = adj[u][i];
82             if (e.cap <= 0) continue;
83             int v = e.to;
84             ll x = e.cost + h[u] - h[v];
85             if (dist[u] != INF && dist[v] > dist[u] + x) {
86                 dist[v] = dist[u] + x;
87                 pv[v] = u;
88                 pe[v] = i;
89                 pq.push({dist[v], v});
90             }
91         }
92     }
93 }
94 };
95 // usage
96 int nodes = 302; // amount of nodes in the network
97 MinCostFlow mcf(nodes);
98 for (int i = 0; i < 150; i++) {
99     mcf.add_edge(0, i+1, 1, 0); // source to node
100     mcf.add_edge(i+151, nodes-1, 1, 0); // node to sink
101 }
102 for (int i = 0; i < n; i++) {
103     int a, b, c; cin >> a >> b >> c;
104     mcf.add_edge(a, b+150, 1, -c); // edges in between (-c to maximize the cost)
105 }
106 // final max cost is -cost
107 auto [cost, flow] = mcf.min_cost_flow(0, nodes-1, 150);

```

3.9 SCC compressing

Condensing a graph into a DAG through its strongly connected components can be useful for DP

```

1 struct SCCCondenser {
2     int n, timer, scc_cnt;
3     vi in, low, scc;

```

```

4     stack<int> st;
5     SCCCondenser(const vvi& adj) {
6         n = adj.size();
7         in.assign(n, 0);
8         low.assign(n, 0);
9         scc.assign(n, -1);
10        timer = scc_cnt = 0;
11        for (int i = 0; i < n; ++i)
12            if (!in[i]) dfs(i, adj);
13    }
14    void dfs(int u, const vvi& adj) {
15        in[u] = low[u] = ++timer;
16        st.push(u);
17        for (int v : adj[u]) {
18            if (!in[v]) {
19                dfs(v, adj);
20                low[u] = min(low[u], low[v]);
21            } else if (scc[v] == -1) {
22                low[u] = min(low[u], in[v]);
23            }
24        }
25        if (low[u] == in[u]) {
26            while (true) {
27                int v = st.top(); st.pop();
28                scc[v] = scc_cnt;
29                if (u == v) break;
30            }
31            scc_cnt++;
32        }
33    }
34    // Returns {DAG, Aggregated Values}
35    pair<vvi, vi> build(const vvi& adj, const vi& val) {
36        vvi dag(scc_cnt);
37        vi scc_val(scc_cnt);
38        set<ii> edges;
39        for (int u = 0; u < n; ++u) {
40            scc_val[scc[u]] += val[u]; // Aggregate values
41            for (int v : adj[u]) {
42                if (scc[u] != scc[v]) {
43                    if (edges.count({scc[u], scc[v]})) continue;
44                    edges.insert({scc[u], scc[v]});
45                    dag[scc[u]].push_back(scc[v]);
46                }
47            }
48        }
49        return {dag, scc_val};
50    }
51 };

```

3.10 SPFA

Detects negative cycles and generates them. If there are negative cycles, it sets the distances for vertices that do not have a shortest path to $-\text{INF}$ Time: $\mathcal{O}(VE)$

```

1 auto spfa = [&](int s, int n) -> pair<vi, vi> {
2     vi dist(n, INF), p(n, -1), cnt(n, 0);
3     vector<bool> inq(n, false), in_neg(n, false);
4     queue<int> q, neg_q;
5     vi cycle;
6     dist[s] = 0;
7     // if you need ANY neg cycle, push all vertices
8     // instead of a fixed source
9     q.push(s);
10    inq[s] = true;
11    while (!q.empty()) {
12        int u = q.front(); q.pop();

```

```

13    inq[u] = false;
14    for (auto [v, w] : adj[u]) {
15        if (dist[v] > dist[u] + w && !in_neg[v]) {
16            dist[v] = dist[u] + w;
17            p[v] = u;
18            cnt[v] = cnt[u] + 1;
19            if (cnt[v] >= n) {
20                if (cycle.empty()) {
21                    int c = v;
22                    for (int i = 0; i < n; i++) c = p[c];
23                    for (int cur = c; ; cur = p[cur]) {
24                        cycle.push_back(cur);
25                        if (cur == c && cycle.size() > 1) break;
26                    }
27                    reverse(all(cycle));
28                }
29                if (!in_neg[v]) {
30                    in_neg[v] = true;
31                    neg_q.push(v);
32                }
33            } else if (!inq[v]) {
34                q.push(v);
35                inq[v] = true;
36            }
37        }
38    }
39 }
40 while (!neg_q.empty()) {
41     int u = neg_q.front(); neg_q.pop();
42     dist[u] = -INF;
43     for (auto [v, w] : adj[u]) {
44         if (!in_neg[v]) {
45             in_neg[v] = true;
46             neg_q.push(v);
47         }
48     }
49 }
50 // dist -INF is there is no shortest path
51 return {dist, cycle};
52 };

```

3.11 Steiner Tree

The Steiner Tree problem finds the minimum-cost tree connecting a given set of **terminals** in a weighted graph. The solution may use non-terminal vertices to reduce the total cost.

- **DP State:** $\text{dp}[\text{mask}][u]$ = min cost to connect the terminals in mask ending at u.
- **Time:** $\mathcal{O}(3^S \cdot N + 2^S \cdot M \log N)$

Requires non-negative edge weights. Problem is NP-hard

```

1 int p = dsu.find(special[0]);
2 for (int i : special) {
3     if (dsu.find(i) != p) {
4         cout << "impossible\n"; return;
5     }
6 }
7 vvi dp(1<<s, vi(n, INF));
8 for (int i = 0; i < s; i++)
9     dp[1<<i][special[i]] = 0;
10 for (int mask = 0; mask < (1<<s); mask++) {

```

```

12 for (int sub = mask; sub > 0; sub = (sub-1)&mask){
13     for (int u = 0; u < n; u++){
14         dp[mask][u] = min(dp[mask][u], dp[sub][u] + dp[
15             mask^sub][u]);
16     }
17 }
18 priority_queue<ii, vii, greater<ii>> pq;
19 for (int u = 0; u < n; u++) pq.push({dp[mask][u], u});
20 while(!pq.empty()){
21     auto [d,u] = pq.top(); pq.pop();
22     if (d > dp[mask][u]) continue;
23     for (auto &[w,v] : adj[u]){
24         if (dp[mask][v] > dp[mask][u] + w) {
25             dp[mask][v] = dp[mask][u] + w;
26             pq.push({dp[mask][v], v});
27         }
28     }
29 }
30 }

```

4 DP

4.1 Bin Packing

Time: $\mathcal{O}(n \cdot 2^n)$ Space: $\mathcal{O}(2^n)$

```

1 vi w(n);
2
3 vector<ii> dp(1<<n, ii(INF,0));
4 // dp[i] = for the subset i(bitmask) (A,B) is the pair
5 // where
6 // A - the min number of knapsacks to store this subset
7 // B - the min size of a used knapsack
8 dp[0] = ii(0,INF);
9 for (int subset = 1; subset < (1<<n); subset++){
10     for (int item = 0; item < n; item++){
11         if (!((subset>>item)&1)) continue;
12         int prevsubset = subset - (1<<item);
13         ii prev = dp[prevsubset];
14
15         if (prev.second + w[item] <= x) {
16             // can fill the knapsack, fill it
17             dp[subset] = min(dp[subset], ii(prev.first, prev.
18                 second+w[item]));
19         } else {
20             // cant fill the knapsack, create a new one
21             dp[subset] = min(dp[subset], ii(prev.first+1, w[
22                 item]));
23         }
24     }
25 }
26 cout << dp[(1<<n)-1].first << endl;

```

4.2 Broken Profile DP

Solves the problem of counting how many ways to fill an $n \times m$ grid using 1×2 tiles. This technique can be used whenever the state dependence is only on the previous state (column). Time: $\mathcal{O}(mn2^n)$ Space: $\mathcal{O}(mn2^n)$

```

1 int dp[1002][12][1024];
2 dp[0][0][0] = 1;
3
4 for (int i = 0; i < m; i++){
5     for (int j = 0; j < n; j++){
6         for (int mask = 0; mask < (1<<n); mask++){
7             if (mask & (1<<j)){
8                 int nxt_mask = mask - (1<<j);
9                 dp[i][j+1][nxt_mask] += dp[i][j][mask];
10                dp[i][j+1][nxt_mask] %= M;
11            } else {
12                int q = mask + (1 << j);
13                dp[i][j+1][q] += dp[i][j][mask];
14                dp[i][j+1][q] %= M;
15                if (j < n-1 && (mask & (1<<(j+1)))==0){
16                    q = mask + (1 << (j+1));
17                    dp[i][j+1][q] += dp[i][j][mask];
18                    dp[i][j+1][q] %= M;
19                }
20            }
21        }
22    }
23 }
24 for (int p = 0; p < (1<<n); p++){
25     dp[i+1][0][p] = dp[i][n][p];
26 }
27 }

```

4.3 Convex Hull Trick (CHT)

• Recurrence form:

$$dp_i = \min_{j < i} (dp_j + (h_i - h_j)^2 + c)$$

Expanding:

$$dp_i = h_i^2 + c + \min_{j < i} ((-2h_j)h_i + h_j^2 + dp_j)$$

Thus, for each j , insert the line

$$f_j(x) = (-2h_j)x + h_j^2 + dp_j$$

and query at $x = h_i$.

- **Slope monotonicity:** If coefficients a_j (slopes) are inserted in strictly decreasing (or increasing) order as j grows, and
- **Query monotonicity:** Values x_i for query come in non-decreasing (min) or increasing (max) order consistent with slope order,
- **Complexity:**

- Insertion + amortized query in $\mathcal{O}(1)$ per operation (pointer walk) under monotonicity.
- Non-monotonic case, generic CHT via binary search: $\mathcal{O}(\log n)$ per query.
- General alternative: Li Chao Tree for insertion-queries in arbitrary order, $\mathcal{O}(\log M)$ per operation (where M is the domain of x).

• Constraints:

- If it cannot be written in linear form, CHT does not apply.
- If there is no monotonicity of slopes or queries, consider Li Chao Tree or CHT variant with binary search.

The example below solves the dp where the recurrence is:

$$dp_i = \min_{j < i} (dp_j + (h_i - h_j)^2 + c).$$

```

1 struct CHT {
2     struct Line { // y = mx + c
3         int m, c;
4         Line(int m, int c) : m(m), c(c) {}
5         int val(int x){
6             return m*x + c;
7         }
8         int floorDiv(int num, int den) {
9             if (den < 0) num = -num, den = -den;
10            if (num >= 0) return num / den;
11            else return - ( (-num + den - 1) / den );
12        }
13        int ceilDiv(int num, int den) {
14            if (den < 0) num = -num, den = -den;
15            if (num >= 0) return (num + den - 1) / den;
16            else return - ( (-num) / den );
17        }
18        int intersect(Line l){
19            // m1x + c1 = m2x + c2
20            // x = (c2 - c1)/(m1 - m2)
21            // if slopes are increasing, use floor div
22            return ceilDiv(1.c - c, m - l.m);
23        }
24    };
25    deque<pair<Line, int>> dq;
26    void insert(int m, int c){
27        Line newLine(m, c);
28        if (!dq.empty() && newLine.m == dq.back().first.m)
29            {
30                // If slopes increasing, change to <=
31                if (newLine.c >= dq.back().first.c) return;
32                else dq.pop_back();
33            }
34        // if slopes increasing, change to <=
35        while (dq.size() > 1 && dq.back().second >= dq.back().first.intersect(newLine)){
36            dq.pop_back();
37        }
38        if (dq.empty()){
39            // assuming queries are positive numbers, may
40            // change to -INF or +INF if needed

```

```

39     dq.emplace_back(newLine, 0);
40     return;
41 }
42 dq.emplace_back(newLine, dq.back().first.intersect(
43     newline));
44 // dont use query and queryNonMonotonicValues in the
45 // same problem
46 int query(int x){
47     while (dq.size() > 1){
48         // if slopes increasing, change to >=
49         if (dq[1].second <= x) dq.pop_front();
50         else break;
51     }
52     return dq[0].first.val(x);
53 }
54 int queryNonMonotonicValues(int x){
55     int l=0, r=dq.size()-1, ans=0;
56     while (l <= r) {
57         int mid = (l+r)>>1;
58         if (dq[mid].second <= x) {
59             ans = mid;
60             l = mid + 1;
61         } else {
62             r = mid - 1;
63         }
64     }
65     return dq[ans].first.val(x);
66 }
67
68 void solve(){
69     int n, c; cin >> n >> c;
70     vi h(n), dp(n);
71     for (auto &x : h) cin >> x;
72     dp[0] = 0;
73     CHT cht;
74     cht.insert(-2*h[0], h[0]*h[0]);
75     for (int i = 1; i < n; i++){
76         dp[i] = cht.query(h[i]) + c + h[i]*h[i];
77         cht.insert(-2*h[i], h[i]*h[i] + dp[i]);
78     }
79     cout << dp[n-1] << endl;
80 }

```

4.4 Edit Distance (Levenshtein)

Very similar to LCS, in the sense that it considers prefixes already computed. Time: $\mathcal{O}(mn)$ Space: $\mathcal{O}(mn)$

```

1 vvi dp(n+1, vi(m+1));
2 for (int i = 0; i <= n; i++) dp[i][0] = i;
3 for (int i = 0; i <= m; i++) dp[0][i] = i;
4 for (int i = 1; i <= n; i++){
5     for (int j = 1; j <= m; j++){
6         dp[i][j] = min(
7             min(dp[i][j-1]+1, dp[i-1][j]+1),
8             dp[i-1][j-1] + (s[i-1]!=t[j-1])
9         );
10    }
11 }

```

4.5 Knapsack - 1D

Time: $\mathcal{O}(nW)$ Space: $\mathcal{O}(W)$

```

1 vi dp(W+1);
2 for (int i = 0; i < n; i++){
3     for (int j = W; j >= w[i]; j--){
4         dp[j] = max(dp[j], v[i] + dp[j-w[i]]);
5     }
6 }

```

4.6 Knapsack - 2D

Time: $\mathcal{O}(nW)$ Space: $\mathcal{O}(nW)$

```

1 vvi dp(n+1, vi(W+1));
2 for (int c = 1; c <= W; c++){
3     for (int i = 1; i <= n; i++){
4         dp[i][c] = dp[i-1][w];
5         if (c-w[i-1] >= 0) {
6             dp[i][c] = max(dp[i][c], dp[i-1][c-w[i-1]] + v[i-1]);
7         }
8     }
9 }

```

4.7 LCS - Longest Common Subsequence

Subsequence generation included here. Time: $\mathcal{O}(mn)$ Space: $\mathcal{O}(mn)$

```

1 vvi dp(n+1, vi(m+1));
2
3 for (int i = 1; i <= n; i++){
4     for (int j = 1; j <= m; j++){
5         if (a[i-1] == b[j-1]) {
6             dp[i][j] = dp[i-1][j-1]+1;
7         } else if (dp[i][j-1] > dp[i-1][j]){
8             dp[i][j] = dp[i][j-1];
9         } else {
10            dp[i][j] = dp[i-1][j];
11        }
12    }
13 }

```

4.8 LiChao Tree

Queries and insertions are all $\mathcal{O}(\log M)$. Where M is the size of the query interval the tree receives.

```

1 // Li Chao tree for min/max over domain [L, R].
2 // For int use eps = 0 and discrete mid+1 split;
3 // For ld use eps>0 and split without +1.
4 template<typename T>
5 struct lichao_tree {
6     // if max lichao, change to ::min()
7     static const T identity = numeric_limits<T>::max();
8     struct Line {
9         T m, c;
10        Line() {
11            m = 0;
12            c = identity;
13        }
14        Line(T m, T c) : m(m), c(c) {}
15        T val(T x) {
16            if (c==identity) return identity;

```

```

17         return m * x + c;
18     }
19 };
20 struct Node {
21     Line line;
22     Node *lc, *rc;
23     Node() : lc(0), rc(0) {}
24 };
25 T L, R, eps;
26 deque<Node> buffer;
27 Node* root;
28 Node* new_node() {
29     buffer.emplace_back();
30     return &buffer.back();
31 }
32 lichao_tree() {}
33 lichao_tree(T _L, T _R, T _eps) {
34     init(_L, _R, _eps);
35 }
36 void clear() {
37     buffer.clear();
38     root = nullptr;
39 }
40 void init(T _L, T _R, T _eps) {
41     clear();
42     L = _L;
43     R = _R;
44     eps = _eps;
45     root = new_node();
46 }
47 void insert(Node* &cur, T l, T r, Line line) {
48     if (!cur) {
49         cur = new_node();
50         cur->line = line;
51         return;
52     }
53     T mid = l + (r-l) / 2;
54     if (r - l <= eps) return;
55     // if max lichao, change to >
56     if (line.val(mid) < cur->line.val(mid))
57         swap(line, cur->line);
58     // if max lichao, change to >
59     if (line.val(l) < cur->line.val(l)) insert(cur->lc,
60         l, mid, line);
61     else insert(cur->rc, mid + 1, r, line);
62 }
63 void insert_range(Node* &cur, int lx, int rx, int l,
64     int r, Line line) {
65     if (rx < l || r < lx || lx > rx || l > r) return;
66     if (!cur) cur = new_node();
67     if (l <= lx && rx <= r) return insert(cur, lx, rx,
68         line);
69     int mid = lx + (rx-lx)/2;
70     insert_range(cur->lc, lx, mid, l, r, line);
71     insert_range(cur->rc, mid+1, rx, l, r, line);
72 }
73 // insert a line that is only valid in range [l,r]
74 void insert_range(int m, int c, int l, int r) {
75     insert_range(root, L, R, l, r, Line(m, c));
76 }
77 T query(Node* &cur, T l, T r, T x) {
78     if (!cur) return identity;
79     T mid = l + (r-l) / 2;
80     T res = cur->line.val(x);
81     if (r - l <= eps) return res;
82     // if max lichao, change min to max
83     if (x <= mid) return min(res, query(cur->lc, l, mid,
84         x));
85     else return min(res, query(cur->rc, mid + 1, r, x));
86 }

```

```

82 }
83 void insert(T m, T c) { insert(root, L, R, Line(m, c)
    ); }
84 T query(T x) { return query(root, L, R, x); }
85 };

```

4.9 LIS - Longest Increasing Subsequence

Time: $\mathcal{O}(n \log n)$

```

1 // dp[i] := min value subseq of len i+1 can end in
2 // dp is increasing sequence -> BS
3 int lis(const vi &a){
4     vi dp;
5     for (int x : a){
6         auto it = lower_bound(all(dp), x);
7         if (it == dp.end()) dp.push_back(x);
8         else *it = x;
9     }
10    return dp.size();
11 }

```

4.10 SOSDP

```

1 int k; // amount of bits
2 vi a(1<<k);
3 // sosdp
4 for (int bit = 0; bit < k; bit++){
5     for (int mask = 0; mask < (1<<k); mask++){
6         if (mask>>bit&1) {
7             a[mask] += a[mask ^ (1<<bit)];
8         }
9     }
10 }
11 // do stuff (such as multiplication for OR convolution)
12 // sosdp inverse
13 for (int bit = 0; bit < k; bit++){
14     for (int mask = 0; mask < (1<<k); mask++){
15         if (mask>>bit&1) {
16             a[mask] -= a[mask ^ (1<<bit)];
17         }
18     }
19 }

```

4.11 Subset Sum

Almost identical to Knapsack, this code contains the subset reconstruction. Time: $\mathcal{O}(nS)$ Space: $\mathcal{O}(nS)$

```

1 vvi dp(n+1, vi(sum+1));
2 vvii p(n+1, vii(sum+1));
3
4 dp[0][0] = 1;
5
6 for (int i = 1; i <= n; i++){
7     for (int s = 1; s <= sum; s++){
8         if (s-a[i-1] >= 0 && dp[i-1][s-a[i-1]]){
9             // sum is possible taking item i
10            p[i][s] = {i-1, s-a[i-1]};
11            dp[i][s] = 1;
12        } else if (dp[i-1][s]) {
13            // sum not possible taking item i
14            // but still possible with other items (<i)

```

```

15            p[i][s] = {i-1, s};
16            dp[i][s] = 1;
17        }
18    }
19 }
20 if (!dp[n][target]) {
21     cout << -1 << endl;
22     return;
23 }
24 vi subset;
25 ii pos = {n, target};
26 while (pos != ii(0,0)){
27     auto [i, s] = pos;
28     if (p[i][s].second != s) subset.push_back(a[i-1]);
29     pos = p[i][s];
30 }

```

4.12 Subset Sum with Bitset

If you only need reachability. Time: $\mathcal{O}(nS/w)$ (where w is sys word size) Space: $\mathcal{O}(S)$

```

1 vi a;
2 bitset<MAXSUM+1> dp; dp[0] = 1;
3 for (int x : a) dp |= (dp << x);

```

5 Trees

5.1 Centroid Decomposition

Recursively splits a tree by its center of mass, building a "Centroid Tree" with a strictly guaranteed $\mathcal{O}(\log N)$ depth. Used for path queries where the tree is unrooted.

Build: $\mathcal{O}(N \log N)$

Sample is solving "count distinct paths with number of edges between k_1 and k_2 "

```

1 struct CentroidDecomposition{
2     vvi t, ct;
3     vi sz, added, cp;
4     int n, root;
5     CentroidDecomposition(const vvi &adj, int _k1, int
        _k2)
6     {
7         t = adj,
8         n = adj.size();
9         cp = sz = added = vi(n);
10        root = -1;
11        ct.resize(n);
12        init(_k1, _k2);
13        decompose(0, -1);
14    }
15    int dfs(int u, int p){
16        sz[u] = 1;
17        for (int v : t[u]){
18            if (v == p || added[v]) continue;
19            sz[u] += dfs(v, u);
20        }
21        return sz[u];

```

```

22    }
23    int centroid(int u, int size, int p = -1){
24        for (int v : t[u]){
25            if (v == p || added[v]) continue;
26            if (2*sz[v] > size) return centroid(v, size, u);
27        }
28        return u;
29    }
30    void decompose(int u, int p){
31        int size = dfs(u, p);
32        int c = centroid(u, size);
33        added[c] = 1;
34
35        if (p == -1) root = c;
36        else ct[p].push_back(c);
37        cp[c] = p;
38
39        process_centroid(c);
40
41        for (int v : t[c])
42            if (!added[v])
43                decompose(v, c);
44    }
45
46    // custom stuff
47    int k1, k2;
48    ll ans;
49    vi freq, distfreq;
50    void init(int _k1, int _k2){
51        ans = 0;
52        k1 = _k1; k2 = _k2;
53        distfreq = freq = vi(k2+1);
54        freq[0] = 1;
55    }
56    void process_centroid(int c){
57        int maxd = 0;
58        int validsum = 0;
59        for (int u : t[c]) if (!added[u]){
60            distfreq[0] = 1;
61            int mxd = 0;
62            auto getdist = [&](auto &&getdist, int u, int p,
                int d) -> void {
63                distfreq[d]++;
64                mxd = max(mxd, d);
65                if (d == k2) return;
66                for (int v : t[u]){
67                    if (v == p || added[v]) continue;
68                    getdist(getdist, v, u, d+1);
69                }
70            }; getdist(getdist, u, c, 1);
71            int sum = validsum;
72            for (int d = 1; d <= mxd; d++){
73                ans += 1ll*sum*distfreq[d];
74                sum -= freq[k2-d];
75                if (k1-d-1 >= 0) sum += freq[k1-d-1];
76            }
77            for (int d = 1; d <= mxd; d++) {
78                if (k1 <= d) validsum += distfreq[d];
79                freq[d] += distfreq[d], distfreq[d] = 0;
80            }
81            maxd = max(maxd, mxd);
82        }
83        for (int i = 1; i <= maxd; i++) freq[i] = 0;
84    }
85 };

```

5.2 Sum of distances - rerooting

Given a tree, $f(u, v) :=$ distance from u to v in the tree, compute

$$\sum_{u, v} f(u, v)$$

. Time: $\mathcal{O}(n)$

```
1 vvi adj;
2 vi sum_down, sum_up, sz;
3 void dfs(int u, int p){
4     for (auto v : adj[u]){
5         if (v == p) continue;
6         dfs(v, u);
7         sz[u] += sz[v];
8         sum_down[u] += sum_down[v];
9     }
10    sum_down[u] += sz[u];
11 }
12 void dfs2(int u, int p, int par_ans){
13     int up_amount = sz[0] - sz[u];
14     sum_up[u] += par_ans + up_amount;
15     int sum = sum_down[u];
16     for (auto v : adj[u]){
17         if (v == p) continue;
18         // new p_ans is going up for is going up for p
19         // + go up to u and go down to other subtrees
20         dfs2(v, u, par_ans + (sz[0] - sz[v]) + sum - (sum_down[v] + sz[v]));
21     }
22 }
```

5.3 Edge HLD

Sometimes the value is on the edges, for this few things need to change, but here is a template. Pre-computation: $\mathcal{O}(n)$ Queries: $\mathcal{O}(\log^2 n)$

```
1 struct EdgeHLD {
2     int n, timer = 0;
3     vvi adj;
4     vi par, dep, sz, heavy, head, val, tin, tout;
5     segtree seg;
6     void init(int _n, vvi& _adj) {
7         n = _n;
8         adj = _adj;
9         par = heavy = vi(n, -1);
10        val = dep = sz = head = tin = tout = vi(n);
11        timer = 0;
12        // edgeWeight[v] = weight of edge (par[v], v), for v > 0
13        // root (0) has no par, so its val is (0)
14        dfs1(0, 0, 0);
15        dfs2(0, 0);
16        vi linear(n);
17        for (int u = 0; u < n; u++)
18            linear[tin[u]] = val[u]; // position stores edge weight
19        seg.init(linear);
20    }
21    int dfs1(int u, int p, int w) {
22        sz[u] = 1;
23        par[u] = p;
```

```
24        val[u] = w;
25        int max_sz = 0;
26        for (auto [v, w] : adj[u]) {
27            if (v == p) continue;
28            dep[v] = dep[u] + 1;
29            int size = dfs1(v, u, w);
30            sz[u] += size;
31            if (size > max_sz) {
32                max_sz = size;
33                heavy[u] = v;
34            }
35        }
36        return sz[u];
37    }
38    void dfs2(int u, int h) {
39        tin[u] = timer++;
40        head[u] = h;
41        if (heavy[u] != -1)
42            dfs2(heavy[u], h);
43        for (auto [v, w] : adj[u]) {
44            if (v != par[u] && v != heavy[u])
45                dfs2(v, v);
46        }
47        tout[u] = timer;
48    }
49    // u must be the child of the edge
50    void update_edge(int u, int val) {
51        seg.set(tin[u], val);
52    }
53    void rangeUpdate(int u, int v, int x) {
54        while (head[u] != head[v]) {
55            if (dep[head[u]] < dep[head[v]]) swap(u, v);
56            seg.rangeUpdate(tin[head[u]], tin[u] + 1, x);
57            u = par[head[u]];
58        }
59        if (dep[u] > dep[v]) swap(u, v);
60        seg.rangeUpdate(tin[u] + 1, tin[v] + 1, x);
61    }
62    void update_subtree(int u, int x) {
63        // updates all in subtree of u (edge p[u] -> u)
64        seg.rangeUpdate(tin[u] + 1, tout[u], x);
65    }
66    segtree::node query(int u, int v) {
67        segtree::node ans;
68        while (head[u] != head[v]) {
69            if (dep[head[u]] < dep[head[v]]) swap(u, v);
70            ans = seg.merge(ans, seg.query(tin[head[u]], tin[u] + 1));
71            u = par[head[u]];
72        }
73        if (dep[u] > dep[v]) swap(u, v);
74        ans = seg.merge(ans, seg.query(tin[u] + 1, tin[v] + 1));
75        return ans;
76    }
77    segtree::node query_subtree(int u) {
78        // query all edges in subtree of u
79        return seg.query(tin[u] + 1, tout[u]);
80    }
81 };
```

5.4 HLD - Heavy light decomposition

If you need to compute a function on a path in a tree and need to support value updates on nodes, HLD is the way. Pre-computation: $\mathcal{O}(n)$ Queries: $\mathcal{O}(\log^2 n)$

OBS: this implementation uses the same segtree as this notebook, with 0-indexing and open-closed interval convention. Ideally, just change the segtree to change the computed function, the HLD struct remains the same. OBS2: this template also supports mass updates (path/subtree) and subtree queries.

```
1 struct HLD {
2     int n, timer = 0;
3     vvi adj;
4     vi par, dep, sz, heavy, head, val, tin, tout;
5     segtree seg;
6     void init(int _n, vvi& _adj, vvi& _val) {
7         n = _n;
8         adj = _adj;
9         val = _val;
10        par = heavy = vi(n, -1);
11        dep = sz = head = tin = tout = vi(n);
12        timer = 0;
13        dfs1(0); dfs2(0, 0);
14        vi linear(n);
15        for (int u = 0; u < n; u++)
16            linear[tin[u]] = val[u];
17        seg.init(linear);
18    }
19    int dfs1(int u) {
20        sz[u] = 1;
21        int max_sz = 0;
22        for (int v : adj[u]) {
23            if (v == par[u]) continue;
24            par[v] = u;
25            dep[v] = dep[u] + 1;
26            int size = dfs1(v);
27            sz[u] += size;
28            if (size > max_sz) {
29                max_sz = size;
30                heavy[u] = v;
31            }
32        }
33        return sz[u];
34    }
35    void dfs2(int u, int h) {
36        tin[u] = timer++;
37        head[u] = h;
38        if (heavy[u] != -1)
39            dfs2(heavy[u], h);
40        for (int v : adj[u]) {
41            if (v != par[u] && v != heavy[u])
42                dfs2(v, v);
43        }
44        tout[u] = timer;
45    }
46    void update(int u, int val) {
47        seg.set(tin[u], val);
48    }
49    void rangeUpdate(int u, int v, int x) {
50        while (head[u] != head[v]) {
51            if (dep[head[u]] < dep[head[v]]) swap(u, v);
52            seg.rangeUpdate(tin[head[u]], tin[u] + 1, x);
53            u = par[head[u]];
54        }
55        if (dep[u] > dep[v]) swap(u, v);
56        seg.rangeUpdate(tin[u], tin[v] + 1, x);
57    }
58    void update_subtree(int u, int x) {
59        seg.rangeUpdate(tin[u], tout[u], x);
60    }
61    segtree::node query(int u, int v) {
```

```

62     segtree::node ans;
63     while (head[u] != head[v]) {
64         if (dep[head[u]] < dep[head[v]])
65             swap(u, v);
66         ans = seg.merge(ans, seg.query(tin[head[u]], tin[
67             u]+1));
68         u = par[head[u]];
69     }
70     if (dep[u] > dep[v]) swap(u, v);
71     ans = seg.merge(ans, seg.query(tin[u], tin[v]+1));
72     return ans;
73 }
74 segtree::node query_subtree(int u){
75     return seg.query(tin[u], tout[u]);
76 }

```

5.5 LCA - RMQ

Pre-computation: $\mathcal{O}(n \log n)$

Queries: $\mathcal{O}(1)$

```

1 struct LCARMQ {
2     SparseTable st;
3     vi tin, dep, et_nodes, et_deps;
4     LCARMQ(vvi &adj){
5         int n = adj.size(), timer = 0;
6         tin = dep = vi(n);
7         et_nodes.reserve(2*n);
8         et_deps.reserve(2*n);
9         auto dfs = [&](auto &&dfs, int u, int p) -> void {
10             et_nodes.push_back(u);
11             et_deps.push_back(dep[u]);
12             tin[u] = timer++;
13             for (int v : adj[u]) if (v != p) {
14                 dep[v] = dep[u]+1;
15                 dfs(dfs, v, u);
16                 et_nodes.push_back(u);
17                 et_deps.push_back(dep[u]);
18             }
19             timer++;
20         };
21         dfs(dfs, 0, 0);
22         st.build(et_deps);
23     }
24     int get(int u, int v){
25         int tu = tin[u], tv = tin[v];
26         if (tu > tv) swap(tu, tv);
27         auto [val, id] = st.min(tu, tv);
28         return et_nodes[id];
29     }
30 };

```

5.6 LCA - binary lifting

Pre-computation: $\mathcal{O}(n \log n)$ Queries: $\mathcal{O}(\log n)$ OBS: just call `dfs(root)` before starting queries.

```

1 vvi adj, up;
2 vi tin, tout;
3 int timer = 0;
4 void dfs(int u, int p){
5     tin[u] = timer++;
6     for (auto v : adj[u]){

```

```

7         if (v == p) continue;
8         up[v][0] = u;
9         for (int dist = 1; dist < LOGN; dist++){
10             up[v][dist] = up[up[v][dist-1]][dist-1];
11         }
12         dfs(v);
13     }
14     tout[u] = timer++;
15 }
16 int isAncestor(int u, int v){
17     return tin[u] <= tin[v] && tout[v] <= tout[u];
18 }
19 int lca(int u, int v){
20     if (isAncestor(u, v)) return u;
21     if (isAncestor(v, u)) return v;
22     for (int dist = LOGN-1; dist >= 0; dist--){
23         if (!isAncestor(up[u][dist], v)) u = up[u][dist];
24     }
25     return up[u][0];
26 }

```

6 Problemas clássicos

6.1 2SAT

Struct for solving 2SAT problems that supports many types of boolean expressions. To add a negated literal use `u`

```

1 // para adicionar negacao usar ~u
2 // Ex: a clausula (a v !b) se traduz para add_or(a, ~b)
3 struct TwoSatSolver {
4     int n;
5     vvi adj, adjT;
6     vector<bool> vis, assignment;
7     vi ts, scc;
8     void build(int _n){
9         n = 2*_n;
10        adj.assign(n, vi());
11        adjT.assign(n, vi());
12    }
13    int get(int u){
14        if (u < 0) return 2*(~u)+1;
15        else return 2*u;
16    }
17    void add_impl(int u, int v){ // u -> v
18        u = get(u), v = get(v);
19        adj[u].push_back(v);
20        adjT[v].push_back(u);
21        adj[v^1].push_back(u^1);
22        adjT[u^1].push_back(v^1);
23    }
24    void add_or(int u, int v){ // u || v
25        add_impl(~u, v);
26    }
27    void add_and(int u, int v){ // u && v
28        add_or(u, u); add_or(v, v);
29    }
30    void add_xor(int u, int v){ // u ~ v (u != v)
31        add_impl(u, ~v);
32        add_impl(~u, v);
33    }
34    void add_equals(int u, int v){ // u == v
35        add_impl(u, v);
36        add_impl(v, u);
37    }
38    void tssort(int u){

```

```

39        vis[u] = true;
40        for (int v : adj[u])
41            if (!vis[v]) tssort(v);
42        ts.push_back(u);
43    }
44    void dfs(int u, int c){
45        scc[u] = c;
46        for (int v : adjT[u])
47            if (!scc[v]) dfs(v, c);
48    }
49    pair<bool, vector<bool>> solve(){
50        ts.clear();
51        vis.assign(n, false);
52        for (int i = 0; i < n; i++)
53            if (!vis[i]) tssort(i);
54        reverse(ts.begin(), ts.end());
55        scc.assign(n, 0);
56        int c = 0;
57        for (int u : ts)
58            if (!scc[u]) dfs(u, ++c);
59        assignment.assign(n/2, false);
60        for (int i = 0; i < n; i += 2){
61            if (scc[i] == scc[i+1]) return {false, {}};
62            assignment[i/2] = scc[i] > scc[i+1];
63        }
64        return {true, assignment};
65    }
66 };

```

6.2 Next Greater Element

One of the classic stack applications. Easy to translate to lower, leq or geq, just change the comparator of the `while`.

```

1 vi ngnt(n, n);
2 stack<ii> st;
3 for (int i = 0; i < n; i++){
4     while (!st.empty() && st.top().first < h[i]){
5         ngnt[st.top().second] = i;
6         st.pop();
7     }
8     st.emplace(h[i], i);
9 }

```

7 Strings

7.1 Aho-Corasick

```

1 const int MAX = 1e6;
2 namespace aho {
3     int next[MAX][26];
4     int link[MAX], sz = 0, freq[MAX], exit[MAX];
5     vi leaves[MAX], order;
6     void insert(const string& pat, int id) {
7         int u = 0;
8         for (char c : pat) {
9             int x = c - 'a';
10            if (!next[u][x]) next[u][x] = ++sz;
11            u = next[u][x];
12        }
13        leaves[u].push_back(id);
14    }
15    void build() {
16        queue<int> q; q.push(0);

```

```

17 link[0] = -1;
18 while (!q.empty()) {
19     int u = q.front(); q.pop();
20     order.push_back(u);
21     for (int x = 0; x < 26; x++) {
22         int v = next[u][x];
23         if (!v) continue;
24         int l = link[u];
25         while (l != -1 && !next[l][x]) l = link[l];
26         link[v] = l == -1 ? 0 : next[l][x];
27         exit[v] = leaves[link[v]].empty() ? exit[link[v]] : link[v];
28         q.push(v);
29     }
30 }
31 }
32 vi query(const string& txt, int num_patterns) {
33     vi ans(num_patterns);
34     int u = 0;
35     for (char c : txt){
36         int x = c - 'a';
37         while (u != -1 && !next[u][x]) u = link[u];
38         u = u == -1 ? 0 : next[u][x];
39         freq[u]++;
40     }
41     for (int i = (int)order.size() - 1; i >= 0; i--) {
42         int u = order[i];
43         if (link[u] != -1) freq[link[u]] += freq[u];
44     }
45     for (int i = 1; i <= sz; i++)
46         for (int id : leaves[i])
47             ans[id] += freq[i];
48     return ans;
49 }
50 }

```

7.2 Hashing

Creation time: $\mathcal{O}(n)$ Access time: $\mathcal{O}(1)$ Space: $\mathcal{O}(n)$

```

1 namespace DoubleHashing {
2     const int MOD1 = 2147483647, MOD2 = 2147483629;
3     using mint1 = mint<MOD1>;
4     using mint2 = mint<MOD2>;
5     vector<mint1> pmod1{1}, invpmod1{1};
6     vector<mint2> pmod2{1}, invpmod2{1};
7     const int MAX_VAL = 256;
8     const int OFFSET = 'a' - 1;
9     mt19937 rng(chrono::steady_clock::now().
10         time_since_epoch().count());
11     const mint1 p1 = uniform_int_distribution<int>(
12         MAX_VAL+1, MOD1-1)(rng);
13     const mint2 p2 = uniform_int_distribution<int>(
14         MAX_VAL+1, MOD2-1)(rng);
15     const mint1 inv1 = mint1::inv(p1);
16     const mint2 inv2 = mint2::inv(p2);
17     void extend(int n){
18         while(pmod1.size() <= n){
19             pmod1.push_back(pmod1.back()*p1);
20             pmod2.push_back(pmod2.back()*p2);
21             invpmod1.push_back(invpmod1.back()*inv1);
22             invpmod2.push_back(invpmod2.back()*inv2);
23         }
24     }
25     struct Hash {
26         mint1 h1;
27         mint2 h2;
28         int len = 0;
29         Hash apply(int x) const {

```

```

27     extend(len+1);
28     return {
29         h1+pmod1[len]*(x-OFFSET),
30         h2+pmod2[len]*(x-OFFSET),
31         len+1
32     };
33 }
34 bool operator<(const Hash &o) const {
35     if (h1.val != o.h1.val) return h1.val < o.h1.val;
36     if (h2.val != o.h2.val) return h2.val < o.h2.val;
37     return len < o.len;
38 }
39 bool operator==(const Hash &o) const {
40     return h1.val == o.h1.val && h2.val == o.h2.val
41         && len == o.len;
42 };
43 struct PrefixHash {
44     vector<Hash> pref;
45     PrefixHash(const string &s){
46         int n = s.length();
47         pref.reserve(n+1);
48         pref.emplace_back();
49         for (auto c : s)
50             pref.push_back(pref.back().apply(c));
51     }
52     Hash get(int l, int r) const {
53         l++, r++;
54         mint1 h1 = (pref[r].h1 - pref[l-1].h1)*invpmod1[l-1];
55         mint2 h2 = (pref[r].h2 - pref[l-1].h2)*invpmod2[l-1];
56         return {h1, h2, r-l+1};
57     }
58 };
59 };

```

7.3 KMP

```

1 vi compute_lps(const string &pat){
2     int m = pat.length();
3     vi lps(m);
4     int len = 0;
5     for (int i = 1; i < m; i++){
6         while(len > 0 && pat[i] != pat[len])
7             len = lps[len-1];
8         if (pat[i] == pat[len]) len++;
9         lps[i] = len;
10    }
11    return lps;
12 }
13 }
14 // find all occurrences (simpler version)
15 vi kmp_search(const string &txt, const string &pat){
16     int n = txt.length(), m = pat.length();
17     vi lps = compute_lps(pat + '#' + txt);
18     vi occurrences;
19     for (int i = 0; i < n+m+1; i++){
20         if (lps[i] == pat.length())
21             occurrences.push_back(i-m*2);
22     }
23     return occurrences;
24 }
25 }
26 vi find_borders(const string &s){
27     vi lps = compute_lps(s);
28     int i = s.length()-1;
29     vi ans;
30     while (lps[i] > 0){

```

```

30     ans.push_back(lps[i]);
31     i = lps[i]-1;
32 }
33 reverse(ans.begin(), ans.end());
34 return ans;
35 }

```

7.4 Manacher

Computes the radius of the longest palindrome centered at every position in linear time, for even sized palindromes, center is the right index.

```

1 // ex:  a b a b b a c a
2 // odd: 1 2 2 1 1 1 2 1
3 // even: 0 0 0 0 2 0 0 0
4 struct Manacher {
5     vi odd, even;
6     int count = 0;
7     Manacher(const string &s) {
8         int n = s.size();
9         odd = even = vi(n);
10        for (int i = 0, l = 0, r = -1; i < n; i++) {
11            int k = i > r ? 1 : min(odd[l+r-i], r-i+1);
12            while (i-k >= 0 && i+k < n && s[i-k] == s[i+k]) k++;
13            odd[i] = k;
14            if (i+k-1 > r) l = i-k+1, r = i+k-1;
15        }
16        for (int i = 0, l = 0, r = -1; i < n; i++) {
17            int k = i > r ? 0 : min(even[l+r-i+1], r-i+1);
18            while (i-k-1 >= 0 && i+k < n && s[i-k-1] == s[i+k]) k++;
19            even[i] = k;
20            if (i+k-1 > r) l = i-k, r = i+k-1;
21        }
22        count = 0;
23        for (int i = 0; i < n; i++) count += odd[i]+even[i];
24    }
25    // query if substring s[l..r] is a palindrome
26    bool is_palindrome(int l, int r) {
27        int len = r-l+1, center = l+len/2;
28        if ((len&1) return odd[center] > len/2;
29        return even[center] >= len/2;
30    }
31 };

```

7.5 Suffix Array

Time: $\mathcal{O}(n \log n)$ Space: $\mathcal{O}(n)$

```

1 struct SuffixArray {
2     vi sa, lcp;
3     SuffixArray(vi s, int lim = 256) {
4         s.push_back(0);
5         int n = s.size(), k = 0, a, b;
6         vi x(all(s)), y(n), ws(max(n, lim));
7         sa = lcp = y, iota(all(sa), 0);
8         for (int j=0, p=0; p<n; j = max(1ll, 2*j), lim=p) {
9             p = j, iota(all(y), n - j);
10            for (int i = 0; i < n; i++)
11                if (sa[i] >= j) y[p++] = sa[i]-j;
12            fill(all(ws), 0);
13            for (int i = 0; i < n; i++) ws[x[i]]++;
14            for (int i = 1; i < lim; i++) ws[i] += ws[i-1];

```

```

15     for (int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
16     swap(x, y), p = 1, x[sa[0]] = 0;
17     for (int i = 1; i < n; i++) {
18         a = sa[i - 1], b = sa[i];
19         x[b] = (y[a]==y[b]&&y[a+j]==y[b+j]) ? p-1:p++;
20     }
21 }
22 for (int i = 0, j; i < n - 1; lcp[x[i++]] = k)
23     for (k&&k--,j=sa[x[i]-1]; s[i+k]==s[j+k]; k++);
24 }
25 };

```

7.6 Suffix Automaton

```

1 #define ALPHABET 26
2 struct SAM {
3     struct State {
4         int len = 0, link = -1;
5         int cnt = 0, first_occ = -1;
6         int next[ALPHABET];
7         State() { fill(next, next + ALPHABET, -1); }
8     };
9     vector<State> st;
10    int last;
11    SAM(const string &s) {
12        st.reserve(2*s.length());
13        st.push_back(State());
14        last = 0;
15        for (int i = 0; i < s.size(); i++) extend(s[i], i);
16        calc_cnt();
17    }
18    void extend(char c, int id) {
19        int cur = st.size();
20        st.push_back(State());
21        st[cur].len = st[last].len + 1;
22        st[cur].first_occ = id;
23        st[cur].cnt = 1;
24        int p = last;
25        while (p != -1 && st[p].next[c-'a'] == -1) {
26            st[p].next[c-'a'] = cur;
27            p = st[p].link;
28        }
29        if (p == -1) {
30            st[cur].link = 0;
31            last = cur;
32            return;
33        }
34        int q = st[p].next[c-'a'];
35        if (st[p].len + 1 == st[q].len) {
36            st[cur].link = q;
37            last = cur;
38            return;
39        }
40        int clone = st.size();
41        st.push_back(st[q]);
42        st[clone].len = st[p].len + 1;
43        st[clone].cnt = 0;
44        while (p != -1 && st[p].next[c-'a'] == q) {
45            st[p].next[c-'a'] = clone;
46            p = st[p].link;
47        }
48        st[q].link = st[cur].link = clone;
49        last = cur;
50    }
51    void calc_cnt() {
52        int sz = st.size();
53        int mx = 0;
54        for (auto &s : st) mx = max(mx, s.len);

```

```

55        vi c(mx+1);
56        vi nodes(sz);
57        for (int i = 0; i < sz; i++) c[st[i].len]++;
58        for (int i = 1; i <= mx; i++) c[i] += c[i-1];
59        for (int i = 0; i < sz; i++) nodes[--c[st[i].len]]
60            = i;
61        for (int i = sz-1; i >= 0; i--) {
62            int u = nodes[i];
63            if (st[u].link != -1) st[st[u].link].cnt += st[u].cnt;
64        }
65    }
66    string longest_common_substring(const string &t) {
67        int v = 0, l = 0, len = 0, pos = 0;
68        for (int i = 0; i < t.length(); i++) {
69            int c = t[i] - 'a';
70            while (v != 0 && st[v].next[c] == -1) {
71                v = st[v].link;
72                l = st[v].len;
73            }
74            if (st[v].next[c] != -1) {
75                v = st[v].next[c];
76                l++;
77            }
78            if (l > len) {
79                len = l;
80                pos = i;
81            }
82        }
83        if (len == 0) return "";
84        return t.substr(pos-len+1, len);
85    }
86    int count_occurrences(const string &t){
87        int cur = 0;
88        for (char c : t){
89            if (st[cur].next[c-'a'] == -1) return 0;
90            cur = st[cur].next[c-'a'];
91        }
92        return st[cur].cnt;
93    }
94    int first_occurrence(const string &t){
95        int cur = 0;
96        for (char c : t){
97            if (st[cur].next[c-'a'] == -1) return -1;
98            cur = st[cur].next[c-'a'];
99        }
100        return st[cur].first_occ-t.length()+1;
101    }
102    ll distinct_substrings(){
103        ll ans = 0;
104        for (int i = 1; i < st.size(); i++)
105            ans += st[i].len - st[st[i].link].len;
106        return ans;
107    }
108    vector<ll> distinct_substrings_perlen(int n){
109        vector<ll> diff(n+2);
110        for (int i = 1; i < st.size(); i++){
111            int l = st[st[i].link].len+1;
112            int r = st[i].len;
113            diff[l]++; diff[r+1]--;
114        }
115        vector<ll> ans(n+1);
116        ans[0] = diff[0];
117        for (int i = 1; i <= n; i++){
118            ans[i] = ans[i-1]+diff[i];
119        }
120        return ans;
121    }
122    vector<ll> dp;
123    void calc_paths(int u){

```

```

123     if (dp[u] != -1) return;
124     dp[u]=1;
125     for (int c = 0; c < ALPHABET; c++){
126         int v = st[u].next[c];
127         if (v == -1) continue;
128         calc_paths(v);
129         dp[u] += dp[v];
130     }
131 }
132 string find_kth(ll k) {
133     dp.assign(st.size(), -1);
134     calc_paths(0);
135     int u = 0;
136     string ans = "";
137     while (k > 0) {
138         for (int c = 0; c < ALPHABET; c++) {
139             int v = st[u].next[c];
140             if (v == -1) continue;
141             if (k <= dp[v]) {
142                 ans += 'a' + c;
143                 u = v;
144                 k--;
145                 break;
146             } else k -= dp[v];
147         }
148     }
149     return ans;
150 }
151 void calc_paths_with_repetitions(int u){
152     if (dp[u] != -1) return;
153     dp[u]=st[u].cnt;
154     for (int c = 0; c < ALPHABET; c++){
155         int v = st[u].next[c];
156         if (v== -1) continue;
157         calc_paths_with_repetitions(v);
158         dp[u] += dp[v];
159     }
160 }
161 string find_kth_with_repetitions(ll k){
162     dp.assign(st.size(), -1);
163     calc_paths_with_repetitions(0);
164     int u = 0;
165     string ans = "";
166     while(k>0){
167         for (int c = 0; c < ALPHABET; c++){
168             int v = st[u].next[c];
169             if (v== -1) continue;
170             if (k <= dp[v]) {
171                 ans += 'a' + c;
172                 u = v;
173                 k -= st[v].cnt;
174                 break;
175             } else k -= dp[v];
176         }
177     }
178     return ans;
179 }
180 };

```

7.7 Z

$$z[i] := \max(k) | s[0..k-1] = s[i..i+k-1]$$

Time: $\mathcal{O}(n + m)$ Space: $\mathcal{O}(n + m)$

```

1 vi compute_z(const string &s) {
2     int n = s.length();
3     vi z(n);
4     int l = 0, r = 0;

```

```

5     for (int i = 1; i < n; i++) {
6         if (i <= r) z[i] = min(r-i+1, z[i-1]);
7         while (i+z[i] < n && s[z[i]] == s[i+z[i]])
8             z[i]++;
9         if (i + z[i] - 1 > r) {
10             l = i; r = i+z[i]-1;
11         }
12     }
13     return z;
14 }
15 vi find_occurrences(const string &txt, const string &
16     pat){
17     vi occurrences;
18     vi z = compute_z(pat + '#' + txt);
19     int n = txt.length(), m = pat.length();
20     for (int i = 0; i < n+m+1; i++){
21         if (z[i] == m) occurrences.push_back(i-m-1);
22     }
23     return occurrences;

```

8 Math

8.1 Combinatorics (Pascal's Triangle)

Computes "n choose k". Requires factorials to be pre-computed. Time: $\mathcal{O}(\log ZAP)$

8.1.1 Combinatorial Analysis

Fundamental Counting Principles

- **Permutations:** The number of ways to arrange k items from a set of n distinct items.

$$P(n, k) = \frac{n!}{(n-k)!}$$

- **Combinations (Binomial Coefficient):** The number of ways to choose k items from a set of n distinct items, regardless of order.

$$\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$$

- **Combinations with Repetition (Stars and Bars):** The number of ways to choose k items of n types, allowing repetitions. Equivalently, the number of ways to distribute k identical balls into n distinct urns.

$$\binom{k+n-1}{n-1} = \binom{k+n-1}{k}$$

Binomial Coefficient Properties and Pascal's Triangle

- **Pascal's Triangle**

$$\begin{array}{c}
 \binom{0}{0} \\
 \binom{1}{0} \binom{1}{1} \\
 \binom{2}{0} \binom{2}{1} \binom{2}{2} \\
 \binom{3}{0} \binom{3}{1} \binom{3}{2} \binom{3}{3} \\
 \binom{4}{0} \binom{4}{1} \binom{4}{2} \binom{4}{3} \binom{4}{4}
 \end{array}$$

- **Stifel's Relation:** Each element in Pascal's Triangle is the sum of the two elements immediately above it.

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

- **Symmetry:** Elements of a row are symmetric with respect to the center. Choosing k elements is the same as choosing the $n-k$ elements to be left behind.

$$\binom{n}{k} = \binom{n}{n-k}$$

- **Row Sum:** The sum of all elements in row n of Pascal's Triangle (where the first row is $n=0$) is equal to 2^n .

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

- **Hockey Stick Identity:** The sum of elements in a diagonal, starting at

$$\binom{r}{r}$$

and ending at

$$\binom{n}{r}$$

, is equal to the element in the next row and next column,

$$\binom{n+1}{r+1}$$

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

- **Binomial Theorem:**

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

- **Vandermonde's Identity:**

$$\sum_{j=0}^k \binom{m}{j} \binom{n}{k-j} = \binom{m+n}{k}$$

The easiest way to understand the identity is through a counting problem. Imagine you have a committee with m men and n women. How many ways can you form a subcommittee of k people?

Way 1 Direct Counting

You have a total of $m+n$ people and need to choose k of them. The number of ways to do this is simply:

$$\binom{m+n}{k}$$

Way 2 Counting by Cases

We can divide the problem into cases, based on how many men (j) are chosen for the subcommittee.

Case 0: Choose 0 men and k women. The number of ways is

$$\binom{m}{0} \binom{n}{k}$$

Case 1: Choose 1 man and $k-1$ women. The number of ways is

$$\binom{m}{1} \binom{n}{k-1}$$

Case j : Choose j men and $k-j$ women. The number of ways is

$$\binom{m}{j} \binom{n}{k-j}$$

Other Important Concepts

- **Catalan Numbers:** A sequence of natural numbers that occurs in various counting problems (e.g., number of binary trees, balanced parenthesis expressions).

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1} \binom{2n}{n}$$

A commonly used combinatorial proof for the Catalan numbers involves counting the number of lattice (grid) paths from $(0,0)$ to (n,n) that do not cross above the diagonal $y = x$. Each such path consists of n rightward steps and n upward steps, and the Catalan number counts the number of these "Dyck paths" that never go above the diagonal.

- **Stirling Numbers of the Second Kind:** The number of ways to partition a set of n labeled objects into k non-empty unlabeled subsets. Denoted by $S(n, k)$ or

$$\{n \atop k\}$$

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

The Stirling numbers of the second kind can also be computed recursively:

$$S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$$

with the boundary conditions:

$$S(0, 0) = 1; \quad S(n, 0) = 0 \text{ for } n > 0; \quad S(0, k) = 0 \text{ for } k > 0$$

- **Bell Number:** The Bell number B^n counts the total number of ways to partition a set of n labeled elements into any number (from 1 up to n) of non-empty, unlabeled subsets. It can also be written as a recurrence relation

$$B^n = \sum_{k=0}^n S(n, k)$$

- **Pigeonhole Principle:** If n items are put into m boxes, with $n > m$, then at least one box must contain more than one item.

8.2 Convolutions

8.2.1 AND convolution

$$c[k] = \sum_{i \& j = k} a[i] \cdot b[j]$$

```
1 vector<mint> and_conv(vector<mint> a, vector<mint> b){
2   int n = a.size(); // must be pow of 2
3   for (int j = 1; j < n; j <= 1) {
4     for (int i = 0; i < n; i++) {
5       if (i&j) {
6         a[i^j] += a[i];
7         b[i^j] += b[i];
8       }
9     }
10  }
11  for (int i = 0; i < n; i++) a[i] *= b[i];
12  for (int j = 1; j < n; j <= 1)
13    for (int i = 0; i < n; i++)
14      if (i&j) a[i^j] -= a[i];
15  return a;
16 }
```

8.2.2 GCD convolution

$$c[k] = \sum_{\gcd(i,j)=k} a[i] \cdot b[j]$$

```
1 vector<mint> gcd_conv(vi a, vi b){
2   int n = (int)max(a.size(), b.size());
3   a.resize(n); b.resize(n);
4   vector<mint> c(n);
5   for (int i = 1; i < n; i++) {
6     mint x = 0;
7     mint y = 0;
8     for (int j = i; j < n; j += i) {
9       x += a[j];
10      y += b[j];
11    }
12    c[i] = x*y;
13  }
14  for (int i = n-1; i >= 1; i--)
15    for (int j = 2*i; j < n; j += i)
16      c[i] -= c[j];
17  return c;
18 }
```

8.2.3 LCM convolution

$$c[k] = \sum_{\text{lcm}(i,j)=k} a[i] \cdot b[j]$$

```
1 vector<mint> lcm_conv(vi a, vi b){
2   int n = (int)max(a.size(), b.size());
3   a.resize(n); b.resize(n);
4   vector<mint> c(n), x(n), y(n);
5   for (int i = 1; i < n; i++) {
6     for (int j = i; j < n; j += i) {
7       x[j] += a[i];
8       y[j] += b[i];
9     }
10    c[i] = x[i]*y[i];
11  }
```

```
12  for (int i = 1; i < n; i++)
13    for (int j = 2 * i; j < n; j += i)
14      c[j] -= c[i];
15  return c;
16 }
```

8.2.4 OR convolution

$$c[k] = \sum_{i|j=k} a[i] \cdot b[j]$$

```
1 vector<mint> or_conv(vector<mint> a, vector<mint> b){
2   int n = a.size(); // must be pow of 2
3   for (int j = 1; j < n; j <= 1) {
4     for (int i = 0; i < n; i++) {
5       if (i&j) {
6         a[i] += a[i^j];
7         b[i] += b[i^j];
8       }
9     }
10  }
11  for (int i = 0; i < n; i++) a[i] *= b[i];
12  for (int j = 1; j < n; j <= 1)
13    for (int i = 0; i < n; i++)
14      if (i&j) a[i] -= a[i^j];
15  return a;
16 }
```

8.2.5 Subset Convolution

$$c[k] = \sum_{i|j=k, i \& j = 0} a[i] \cdot b[j]$$

Complexity: $\mathcal{O}(k^2 2^k)$, where k is the number of bits.

```
1 vector<mint> subset_conv(const vector<mint>& a, const
2   vector<mint>& b) {
3   int n = a.size(); // must be pow of 2
4   int k = __builtin_ctz(n);
5   vector<vector<mint>> fa(k+1, vector<mint>(n, 0));
6   vector<vector<mint>> fb(k+1, vector<mint>(n, 0));
7   vector<vector<mint>> fc(k+1, vector<mint>(n, 0));
8   for (int i = 0; i < n; i++) {
9     int pc = __builtin_popcount(i);
10    fa[pc][i] = a[i];
11    fb[pc][i] = b[i];
12  }
13  for (int pc = 0; pc <= k; pc++) {
14    for (int j = 1; j < n; j <= 1) {
15      for (int i = 0; i < n; i++) {
16        if (i & j) {
17          fa[pc][i] += fa[pc][i ^ j];
18          fb[pc][i] += fb[pc][i ^ j];
19        }
20      }
21    }
22  }
23  for (int i = 0; i < n; i++) {
24    for (int pc = 0; pc <= k; pc++) {
25      for (int j = 0; j <= pc; j++) {
26        fc[pc][i] += fa[j][i] * fb[pc - j][i];
27      }
28    }
29  }
```

```

27 }
28 }
29 for (int pc = 0; pc <= k; pc++) {
30     for (int j = 1; j < n; j <= 1) {
31         for (int i = 0; i < n; i++) {
32             if (i & j) fc[pc][i] -= fc[pc][i ^ j];
33         }
34     }
35 }
36 vector<mint> c(n);
37 for (int i = 0; i < n; i++) {
38     c[i] = fc[___builtin_popcount(i)][i];
39 }
40 return c;
41 }

```

8.2.6 XOR convolution

$$c[k] = \sum_{i \oplus j = k} a[i] \cdot b[j]$$

```

1 void fwht(vector<mint> &a, bool inv){
2     int n = a.size(); // must be pow of 2
3     for (int step = 1; step < n; step <= 1){
4         for (int i = 0; i < n; i += 2*step) {
5             for (int j = i; j < i+step; j++){
6                 auto u = a[j];
7                 auto v = a[j+step];
8                 a[j] = u+v;
9                 a[j+step] = u-v;
10            }
11        }
12    }
13    if (inv) for (auto &x : a) x /= n;
14 }
15 vector<mint> xor_conv(vector<mint> a, vector<mint> b){
16     int n = a.size();
17     fwht(a,0), fwht(b, 0);
18     for (int i = 0; i < n; i++) a[i] *= b[i];
19     fwht(a,1);
20     return a;
21 }

```

8.3 Extended Euclid

Time: $\mathcal{O}(\log n)$.

```

1 int extended_gcd(int a, int b, int &x, int &y) {
2     x = 1, y = 0;
3     int x1 = 0, y1 = 1;
4     while (b) {
5         int q = a / b;
6         tie(x, x1) = make_tuple(x1, x - q * x1);
7         tie(y, y1) = make_tuple(y1, y - q * y1);
8         tie(a, b) = make_tuple(b, a - q * b);
9     }
10    return a;
11 }

```

8.4 Factorization

Fast factorization using Pollard-Rho Time: $\mathcal{O}(n^{1/4})$

```

1 void factorize(u64 n, vector<u64>& factors) {
2     if (n <= 1) return;
3     if (miller_rabin(n)) {
4         factors.push_back(n);
5         return;
6     }
7     u64 divisor = pollard_rho(n);
8     factorize(divisor, factors);
9     factorize(n / divisor, factors);
10 }

```

8.5 Gaussian elimination

Solves system of n linear equations for m variables, or informs there is no solution. Template is adaptable to use mint if necessary. a should have the last column as the "b" values. Time: $\mathcal{O}(n^2m)$

```

1 pair<int, vector<ld>> gauss(vector<vector<ld>> a) {
2     const double eps = 1e-6;
3     int n = a.size(), m = a[0].size()-1;
4     vi where(m, -1);
5     for (int var = 0, eq = 0; var < m && eq < n; var++) {
6         int piv = eq;
7         for (int i=eq; i<n; ++i)
8             if (abs(a[i][var]) > abs(a[piv][var])) piv = i;
9         if (abs(a[piv][var]) < eps) continue;
10        for (int i = var; i <= m; i++)
11            swap(a[piv][i], a[eq][i]);
12        where[var] = eq;
13        for (int i = 0; i < n; i++) if (i != eq) {
14            ld c = a[i][var] / a[eq][var];
15            for (int j = var; j <= m; j++)
16                a[i][j] -= a[eq][j] * c;
17        }
18        eq++;
19    }
20
21    vector<ld> ans(m, 0);
22    for (int i = 0; i < m; i++) if (where[i] != -1)
23        ans[i] = a[where[i]][m] / a[where[i]][i];
24    for (int i = 0; i < n; i++) {
25        ld sum = 0;
26        for (int j = 0; j < m; j++)
27            sum += ans[j] * a[i][j];
28        if (abs(sum - a[i][m]) > eps) return {0, {}};
29    }
30
31    for (int i = 0; i < m; i++) if (where[i] == -1)
32        return {INF, ans};
33    return {1, ans};
34 }

```

8.6 Inclusion-Exclusion Principle

For a set of k primes, count numbers from 1 to n divisible by at least one prime.

For every non-empty subset of primes:

- If the subset has odd size, add $n/prod$.
- If the subset has even size, subtract $n/prod$.
- $prod$ is the product of the selected primes.

Here, $n/prod$ is the number of integers from 1 to n divisible by all selected primes.

Complexity: $\mathcal{O}(2^k k)$.

```

1 int n, ans = 0;
2 vi primes;
3 for (int mask = 1; mask < (1<<primes.size()); mask++){
4     int prod = 1;
5     for (int bit = 0; bit < primes.size(); bit++) {
6         if (mask>>bit&1) {
7             prod *= primes[bit];
8             if (prod > n) break;
9         }
10    }
11    if (___popcount(mask) & 1) ans += n/prod;
12    else ans -= n/prod;
13 }

```

8.7 Legendre's formula

Computes the largest power of a prime p in $n!$

```

1 int legendre(int n, int p){
2     int ans = 0;
3     while(n>0){ n/=p; ans += n; }
4     return ans;
5 }

```

8.8 Linear Sieve + Möbius Function

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1 \\ (-1)^k & \text{if } n \text{ is a product of } k \text{ distinct primes} \\ 0 & \text{if } p^2 \mid n \text{ for some prime } p \end{cases}$$

8.8.1 Applications

- **Inclusion-Exclusion:** Naturally generates $1, -1, 0$ coefficients for prime factor sets.
- **Möbius Inversion:** Converts $g(n) = \sum_{d|n} f(d)$ to $f(n) = \sum_{d|n} \mu(d)g(\frac{n}{d})$. Often reduces $\mathcal{O}(N)$ divisor sums to $\mathcal{O}(\sqrt{N})$.
- **Dirichlet Convolution:** $\mu * 1 = \epsilon$. Core to sub-linear prefix sum algorithms (e.g., DuSieve) yielding $\mathcal{O}(N^{2/3})$ time.

```

1 auto sieve = [](int n) -> pair<vi,vi> {
2     vi primes, mu(n+1);
3     mu[1] = 1;
4     vector<bool> is_prime(n+1, true);
5     is_prime[0] = is_prime[1] = false;
6     for (int i = 2; i <= n; ++i) {
7         if (is_prime[i]) {
8             primes.push_back(i);
9             mu[i] = -1;
10        }
11        for (int p : primes) {
12            if (i*p > n) break;
13            is_prime[i*p] = false;
14            if (i%p == 0) {
15                mu[i*p] = 0;
16                break;
17            }
18            mu[i*p] = -mu[i];
19        }
20    }
21    return {primes, mu};
22 };

```

8.9 Matrix template

A template for square matrices, used for solving linear recurrences with fast exponentiation, memory is on a 1D vector, but can be accessed with `A[i][j]` normally (custom `operator[]`)

```

1 template<typename T>
2 struct mat{
3     vector<T> m;
4     int n;
5     mat(int _n = 0, bool identity = false) : n(_n) {
6         m.resize(n*n);
7         if (!identity) return;
8         for (int i = 0; i < n; i++)
9             m[i*n+i] = 1;
10    }
11    mat& operator+=(const mat &o){
12        for (int i = 0; i < n; i++){
13            int ra = i*n;
14            for (int j = 0; j < n; j++){
15                m[ra+j] += o.m[ra+j];
16            }
17        }
18        return *this;
19    }
20    mat& operator-=(const mat& o){
21        for (int i = 0; i < n; i++){
22            int ra = i*n;
23            for (int j = 0; j < n; j++){
24                m[ra+j] -= o.m[ra+j];
25            }
26        }
27        return *this;
28    }
29    mat& operator*=(const mat& o){
30        vector<T> ans(n*n);
31        for (int i = 0; i < n; i++){
32            for (int k = 0; k < n; k++){
33                int ra = i*n, rb = k*n;
34                for (int j = 0; j < n; j++){
35                    ans[ra+j] += m[ra+k] * o.m[rb+j];
36                }
37            }
38        }
39    }

```

```

37    }
38    }
39    this->m = ans;
40    return *this;
41 }
42 friend mat operator+(mat a, const mat& b){return a+=b;
43 }
44 friend mat operator-(mat a, const mat& b){return a-=b;
45 }
46 friend mat operator*(mat a, const mat& b){return a*=b;
47 }
48 T* operator[](int i){
49     return &m[i*n];
50 }
51 vector<T> operator*(const vector<T> &v){
52     vector<T> ans(n);
53     for (int i = 0; i < n; i++){
54         int ra = i*n;
55         for (int j = 0; j < n; j++){
56             ans[i] += m[ra+j]*v[j];
57         }
58     }
59     return ans;
60 }
61 static mat exp(mat b, ll e){
62     mat ans = mat(b.n,true);
63     while(e>0){
64         if(e&1) ans*=b;
65         b*=b;
66         e>>=1;
67     }
68     return ans;
69 }

```

8.10 Miller-Rabin primality test

Time: $\mathcal{O}(k \log n)$ (k is the amount of bases tested, i.e. 7)

```

1 using u64 = uint64_t;
2 using u128 = __uint128_t;
3 u64 modpow(u64 b, u64 e, u64 mod) {
4     u64 ans = 1;
5     for (b %= mod; e; e >>= 1, b = (u128)b * b % mod)
6         if (e & 1) ans = (u128)ans * b % mod;
7     return ans;
8 }
9 // if up to 32 bits, use {2, 7, 61}
10 u64 primes[] = {2, 325, 9375, 28178, 450775, 9780504,
11                1795265022};
12 bool miller_rabin(u64 n) {
13     if (n < 2) return false;
14     if (n % 2 == 0) return n == 2;
15     u64 d = n - 1;
16     int s = 0;
17     while (d % 2 == 0) d >>= 1, s++;
18     for (u64 a : primes) {
19         if (a%n == 0) continue;
20         u64 x = modpow(a, d, n);
21         if (x == 1 || x == n - 1) continue;
22         bool composite = true;
23         for (int r = 1; r < s; r++) {
24             x = (u128)x * x % n;
25             if (x == n - 1) { composite = false; break; }
26         }
27         if (composite) return false;
28     }
29     return true;
30 }

```

8.11 Mint

```

1 template<const int MOD>
2 struct mint {
3     using m = mint;
4     int val;
5     m(ll v=0){
6         ll x = v%MOD;
7         if (x<0) x+=MOD;
8         val = x;
9     }
10    m& operator+=(const m &o){
11        ll x = (ll)val + o.val;
12        if (x >= MOD) x -= MOD;
13        val = x;
14        return *this;
15    }
16    m& operator-=(const m &o){
17        ll x = (ll)val - o.val;
18        if (x < 0) x += MOD;
19        val = x;
20        return *this;
21    }
22    m& operator*=(const m &o){
23        val = (ll)val * o.val % MOD;
24        return *this;
25    }
26    m& operator/=(const m &o){
27        val = (ll)val * inv(o).val % MOD;
28        return *this;
29    }
30    friend m operator+(m a, const m &b) {return a+=b;}
31    friend m operator-(m a, const m &b) {return a-=b;}
32    friend m operator*(m a, const m &b) {return a*=b;}
33    friend m operator/(m a, const m &b) {return a/=b;}
34    static m power(m b, int e){
35        m ans = 1;
36        while(e>0){
37            if (e&1) ans *= b;
38            b*=b; e>>=1;
39        }
40        return ans;
41    }
42    static m inv(m n){ return power(n,MOD-2); }
43 };

```

8.12 Modular Inverse

If m is prime, can use binary exponentiation to compute a^{p-2} (Fermat's Little Theorem).

This code works for non-prime m , as long as it is coprime to a .

Time: $\mathcal{O}(\log m)$

```

1 int modInverse(int a, int m) {
2     int x, y;
3     int g = extendedGcd(a, m, x, y);
4     if (g != 1) return -1;
5     return (x % m + m) % m;
6 }

```

8.13 Number Theoretic Transform (NTT)

NTT is a fast algorithm (analogous to FFT) for polynomial multiplication modulo a special prime. It requires a prime modulus $p = c \cdot 2^k + 1$ (a "primitive root prime") and a primitive 2^k -th root of unity modulo p .

- **Prime Choices:** To use NTT, pick a modulus and a matching primitive root (see table below). For arbitrary moduli (e.g., $10^9 + 7$), multiply with several NTT-friendly primes and reconstruct with CRT (see `crt_multiply`).
- **Time Complexity:** $\mathcal{O}(n \log n)$ for polynomial multiplication.

8.13.1 NTT-Friendly Primes and Roots

NTT-friendly primes and their primitive roots:

- Mod: 998244353, Root: 3, Max N: 2^{23}
- Mod: 734003201, Root: 3, Max N: 2^{20}
- Mod: 167772161, Root: 3, Max N: 2^{25}
- Mod: 469762049, Root: 3, Max N: 2^{26}

Use the modulus as MOD and the root as ROOT when instantiating the NTT.

- For large/concrete moduli, see `crt_multiply` in the code for a multi-modulus solution with Chinese Remainder Theorem (CRT).

```
1 namespace NTT{
2   template<typename T, ll MOD, ll ROOT>
3   void transform(vector<T>& a, bool invert) {
4     int n = a.size();
5
6     for (int i = 1, j = 0; i < n; i++) {
7       int bit = n >> 1;
8       for (; j & bit; bit >>= 1) j ^= bit;
9       j ^= bit;
10      if (i < j) swap(a[i], a[j]);
11    }
12
13    for (int len = 2; len <= n; len <= 1) {
14      T wlen = T::power(ROOT, (MOD - 1) / len);
15      if (invert) wlen = T::inv(wlen);
16      for (int i = 0; i < n; i += len) {
17        T w = 1;
18        for (int j = 0; j < len / 2; j++) {
19          T u = a[i + j], v = a[i + j + len / 2] * w;
20          a[i + j] = u + v;
21          a[i + j + len / 2] = u - v;
22          w *= wlen;
23        }
24      }
25    }
26
27    if (invert) {
28      T n_inv = T::inv(n);
```

```
29      for (T& x : a) x *= n_inv;
30    }
31  }
32
33  template<typename T, ll MOD, ll ROOT>
34  vector<ll> multiply(const vector<ll>& a, const vector<ll>& b) {
35    vector<T> fa(a.begin(), a.end()), fb(b.begin(), b.end());
36    int n = 1;
37    while (n < a.size() + b.size()) n <= 1;
38    fa.resize(n);
39    fb.resize(n);
40
41    transform<T, MOD, ROOT>(fa, false);
42    transform<T, MOD, ROOT>(fb, false);
43
44    for (int i = 0; i < n; i++) fa[i] *= fb[i];
45
46    transform<T, MOD, ROOT>(fa, true);
47
48    vector<ll> result(n);
49    for (int i = 0; i < n; i++) result[i] = fa[i].val;
50    return result;
51  }
52
53  vector<ll> crt_multiply(const vector<ll>& a, const vector<ll>& b) {
54    const ll mod1 = 998244353;
55    const ll root1 = 3;
56    using mint1 = mint<mod1>;
57    vector<ll> ans1 = NTT::multiply<mint1, mod1, root1>(a, b);
58
59    const ll mod2 = 1004535809;
60    const ll root2 = 3;
61    using mint2 = mint<mod2>;
62    vector<ll> ans2 = NTT::multiply<mint2, mod2, root2>(a, b);
63
64    int ans_size = a.size() + b.size() - 2;
65    ll M1_inv_M2 = mint<mod2>::inv(mod1).val;
66
67    vector<ll> final_result(ans_size + 1);
68    for (int i = 0; i <= ans_size; ++i) {
69      ll v1 = ans1[i];
70      ll v2 = ans2[i];
71      ll k = ((v2 - v1 + mod2) % mod2 * M1_inv_M2) % mod2;
72      final_result[i] = v1 + k * mod1;
73    }
74    return final_result;
75  }
76 }
```

8.14 Pollard-Rho

Returns a non-trivial factor of n , used for fast factorization
Time: $\mathcal{O}(n^{1/4})$

```
1 u64 pollard_rho(u64 n) {
2   if (n % 2 == 0) return 2;
3   if (miller_rabin(n)) return n;
4   static mt19937_64 rng(1337); // change to chrono
5   u64 x = 0, y = 0, c = 0, prd = 1, d = 1;
6   auto f = [&](u64 v) { return ((u128)v*v + c)%n; };
7   while (d == 1 || d == n) {
```

```
8     x = y = rng() % (n - 2) + 2;
9     c = rng() % (n - 1) + 1;
10    d = 1;
11    for (int step = 1; d == 1; step <= 1) {
12      x = y;
13      for (int i = 0; i < step; ++i) y = f(y);
14      for (int k = 0; k < step && d == 1; k += 128) {
15        prd = 1;
16        int limit = min(step - k, 128);
17        for (int i = 0; i < limit; ++i) {
18          y = f(y);
19          prd = (u128)prd * (x > y ? x - y : y - x)%n;
20        }
21        d = gcd(prd, n);
22      }
23    }
24  }
25  return d;
26 }
```

8.15 Euler's Totient

Returns the amount of numbers smaller than n that are coprime to n . Time: $\mathcal{O}(\sqrt{n})$

```
1 int phi(int n){
2   int ans = n;
3   for (int i = 2; i*i <= n; i++){
4     if (n%i == 0){
5       while(n%i == 0) n/=i;
6       ans -= ans/i;
7     }
8   }
9   if (n>1) ans -= ans/n;
10  return ans;
11 }
```

8.16 Xor Basis

Powerful for some weird xor problems, represents the linearly independent basis whose span is the set of elements you inserted into it.

```
1 #define D 60 // max bits
2 struct XorBasis {
3   ll a[D] = {0};
4   int sz = 0;
5   // Returns {represented_part, orthogonal_part}
6   pair<ll, ll> reduce(ll x) {
7     ll orig = x;
8     for (int i = D-1; i >= 0; i--)
9       if ((x>>i&1) && a[i]) x ^= a[i];
10    return {orig^x, x};
11  }
12  bool insert(ll x) {
13    for (int i = D-1; i >= 0; i--) {
14      if (!(x>>i&1)) continue;
15      if (!a[i]) {
16        a[i] = x; sz++;
17        return true;
18      }
19      x ^= a[i];
20    }
21    return false;
```

```

22 }
23 bool check(ll x) { return reduce(x).second == 0; }
24 ll max_xor(ll val = 0) {
25     for (int i = D-1; i >= 0; i--)
26         val = max(val, val^a[i]);
27     return val;
28 }
29 ll span_size() { return ll1 << sz; }
30 };

```

9 Geometry

9.1 Geometry

Always try to use integer coordinates. And avoid using dist for everything (hypot is slow AF), use dist2.

```

1 // heavily inspired by brunomaletta/biblioteca
2 using T = long long; // change to ld if necessary
3 const ld DINF = 2e18;
4 const ld pi = acos(-1.0);
5 const ld eps = 1e-9;
6 #define sq(x) ((x)*(x))
7 // -1 for negative, 0 for zero, 1 for positive
8 int sgn(T x) {
9     if constexpr (is_floating_point_v<T>)
10         return (x > eps) - (x < -eps);
11     return (x > 0) - (x < 0);
12 }
13 struct pt {
14     T x, y;
15     pt(T x_ = 0, T y_ = 0) : x(x_), y(y_) {}
16     bool operator<(const pt p) const {
17         if (sgn(x-p.x) != 0) return sgn(x-p.x) < 0;
18         return sgn(y-p.y) < 0;
19     }
20     bool operator==(const pt p) const {
21         return sgn(x-p.x) == 0 && sgn(y-p.y) == 0;
22     }
23     pt operator+(pt p) const {return pt(x+p.x, y+p.y);}
24     pt operator-(pt p) const {return pt(x-p.x, y-p.y);}
25     pt operator*(T c) const {return pt(x*c, y*c);}
26     pt operator/(T c) const {return pt(x/c, y/c);}
27     T operator*(pt p) const {return x*p.x + y*p.y;}
28     T operator^(pt p) const {return x*p.y - y*p.x;}
29     friend istream& operator >> (istream& in, pt& p) {
30         return in >> p.x >> p.y;
31     }
32 };
33 struct line {
34     pt p, q;
35     line() {}
36     line(pt p_, pt q_) : p(p_), q(q_) {}
37     friend istream& operator >> (istream& in, line& r) {
38         return in >> r.p >> r.q;
39     }
40 };
41
42 // --- INTEGER SAFE OPERATIONS --- //
43
44 ld dist(pt p, pt q) {
45     return hypot((ld)(p.y - q.y), (ld)(p.x - q.x));
46 }
47 T dist2(pt p, pt q) {return sq(p.x-q.x) + sq(p.y-q.y);}
48 ld norm(pt v) { return dist(pt(0, 0), v); }
49 // Angle with +x axis in [0, 2*pi)

```

```

50 ld angle(pt v) {
51     ld ang = atan2((ld)v.y, (ld)v.x);
52     return ang < 0 ? ang + 2*pi : ang;
53 }
54 // 2x signed area of triangle p-q-r (>0 if ccw)
55 T sarea2(pt p, pt q, pt r) { return (q-p)^(r-q); }
56 // True if p, q, r are collinear
57 bool col(pt p, pt q, pt r) {return sgn(sarea2(p,q,r))==0;}
58 // True if r is strictly to the left of line p->q
59 bool ccw(pt p, pt q, pt r) {return sgn(sarea2(p,q,r))>0;}
60 bool isvertical(line r) {return sgn(r.p.x-r.q.x) == 0;}
61 // True if point p lies on segment r
62 bool isinseg(pt p, line r) {
63     pt a = r.p - p, b = r.q - p;
64     return sgn(a^b) == 0 && sgn(a*b) <= 0;
65 }
66 bool interseg(line r, line s) {
67     if (isinseg(r.p, s) || isinseg(r.q, s) ||
68         isinseg(s.p, r) || isinseg(s.q, r)) return 1;
69     return ccw(r.p, r.q, s.p) != ccw(r.p, r.q, s.q) &&
70         ccw(s.p, s.q, r.p) != ccw(s.p, s.q, r.q);
71 }
72 ld disttoline(pt p, line r) {
73     return (ld)abs(sarea2(p, r.p, r.q)) / dist(r.p, r.q);
74 }
75 ld disttoseg(pt p, line r) {
76     if (sgn((r.q-r.p)*(p-r.p)) < 0) return dist(r.p,p);
77     if (sgn((r.p-r.q)*(p-r.q)) < 0) return dist(r,q,p);
78     return disttoline(p, r);
79 }
80
81 // --- POLYGONS (INTEGER SAFE) --- //
82
83 ld polarea(const vector<pt> &v) {
84     T ans = 0;
85     for (int i = 0; i < sz(v); i++)
86         ans += (v[i]^v[(i+1)%sz(v)]);
87     return abs((ld)ans / 2.0);
88 }
89 // Returns 0 if out, 1 if inside, 2 if on border
90 int inpol(const vector<pt> &v, pt p) {
91     int qt = 0;
92     for (int i = 0; i < sz(v); i++) {
93         if (p == v[i]) return 2;
94         int j = (i+1)%sz(v);
95         if (sgn(p.y-v[i].y)==0 && sgn(p.y-v[j].y)==0) {
96             if (sgn((v[i]-p)*(v[j]-p)) <= 0) return 2;
97             continue;
98         }
99         bool lower = sgn(v[i].y - p.y) < 0;
100         if (lower == (sgn(v[j].y - p.y) < 0)) continue;
101         auto t = (p-v[i])^(v[j]-v[i]);
102         if (sgn(t) == 0) return 2;
103         if (lower == (sgn(t) > 0)) qt += lower ? 1 : -1;
104     }
105     return qt != 0;
106 }
107 // Monotone chain convex hull - O(N log N)
108 // icol - include_collinear
109 vector<pt> convex_hull(vector<pt> v, bool icol = 0) {
110     sort(all(v));
111     v.erase(unique(all(v)), v.end());
112     int n = sz(v);
113     if (n <= 1) return v;
114     vector<pt> l, u;
115     auto bad = [&](pt a, pt b, pt c) {
116         if (icol) return sgn(sarea2(a, b, c)) < 0;
117         return sgn(sarea2(a, b, c)) <= 0;
118     };
119     for (int i = 0; i < n; i++) {

```

```

120         while (sz(l)>1 &&
121             bad(l[sz(l)-2], l[sz(l)-1], v[i]))
122             l.pop_back();
123         l.push_back(v[i]);
124     }
125     for (int i = n-1; i >= 0; i--) {
126         while (sz(u)>1 && bad(u[sz(u)-2], u[sz(u)-1], v[i]))
127             u.pop_back();
128         u.push_back(v[i]);
129     }
130     l.pop_back(); u.pop_back();
131     l.insert(l.end(), all(u));
132     return l;
133 }
134 // d^2 of closest pair - O(N log N)
135 T closest_pair(vector<pt> v) {
136     if (sz(v) <= 1) return 0;
137     sort(all(v));
138     vector<pt> t(sz(v));
139     auto go = [&](auto& go, int l, int r) -> T {
140         if (r-l <= 1) return DINF;
141         int m = l + (r-l)/2;
142         T mid = v[m].x;
143         T ans = min(go(go, l, m), go(go, m, r));
144         merge(begin(v)+l, begin(v)+m, begin(v)+m, begin(v)+
145             r,
146             begin(t), [](pt a, pt b){return sgn(a.y-b.y)
147                 < 0;});
148         copy(t.begin(), t.begin()+(r-l), v.begin()+l);
149         int k = 0;
150         for (int i = l; i < r; i++) {
151             if (sgn(sq(v[i].x - mid) - ans) >= 0) continue;
152             for (int j = k-1; j >= 0; j--) {
153                 if (sgn(sq(v[i].y-t[j].y)-ans) >= 0) break;
154                 ans = min(ans, dist2(v[i], t[j]));
155             }
156             t[k++] = v[i];
157         }
158         return ans;
159     };
160     return go(go, 0, sz(v));
161 }
162 struct convex_pol {
163     vector<pt> pol;
164     convex_pol(vector<pt> v) : pol(convex_hull(v)) {}
165     // O(log N) check if p is inside convex polygon
166     bool is_inside(pt p) {
167         if (pol.empty()) return false;
168         if (sz(pol) == 1) return p == pol[0];
169         int l = 1, r = sz(pol);
170         while (l < r) {
171             int m = l + (r - l) / 2;
172             if (ccw(p, pol[0], pol[m])) l = m+1;
173             else r = m;
174         }
175         if (l == 1) return isinseg(p, line(pol[0], pol[1]));
176         if (l == (int)sz(pol)) return false;
177         return !ccw(p, pol[l], pol[l-1]);
178     }
179     // O(log N) extreme vertex given custom comparator
180     int extreme(const function<bool(pt, pt)>& cmp) {
181         int n = sz(pol);
182         auto extr = [&](int i, bool& cur_dir) {
183             cur_dir = cmp(pol[(i+1)%n], pol[i]);
184             return !cur_dir && !cmp(pol[(i+n-1)%n], pol[i]);
185         };
186         bool last_dir, cur_dir;
187         if (extr(0, last_dir)) return 0;

```

```

187 int l = 0, r = n;
188 while (l+1 < r) {
189     int m = l + (r - l) / 2;
190     if (extr(m, cur_dir)) return m;
191     bool rel_dir = cmp(pol[m], pol[l]);
192     if ((!last_dir && cur_dir) ||
193         (last_dir==cur_dir && rel_dir==cur_dir)) {
194         l = m; last_dir = cur_dir;
195     } else r = m;
196 }
197 return l;
198 }
199 // O(log N) furthest vertex in direction of vector v
200 int max_dot(pt v) {
201     return extreme(
202         [&](pt p, pt q) { return sgn(p*v - q*v) > 0; }
203     );
204 }
205 // O(log N) indices of tangent vertices
206 // from point p strictly outside
207 pair<int, int> tangents(pt p) {
208     return {
209         extreme([&](pt q, pt r){ return ccw(p,r,q); })),
210         extreme([&](pt q, pt r){ return ccw(p,q,r); }));
211 }
212 };
213
214 // =====//
215 // --- FLOAT ONLY (REQUIRE 'using T = ld;') --- //
216 // =====//
217
218 // th in radians
219 pt rotate(pt p, ld th) {
220     return pt(
221         p.x*cos(th) - p.y*sin(th),
222         p.x*sin(th) + p.y*cos(th)
223     );
224 }
225 pt rotate90(pt p) { return pt(-p.y, p.x); }
226 // Scalar t such that
227 // r.p + t*(r.q - r.p) intersects line pt(0,0) -> v
228 ld get_t(pt v, line r) {
229     return (ld)(r.p-r.q) / (ld)((r.p-r.q)^v);
230 }
231 pt proj(pt p, line r) {
232     if (r.p == r.q) return r.p;
233     r.q = r.q - r.p; p = p - r.p;
234     return r.q * ((p*r.q) / (r.q*r.q)) + r.p;
235 }
236 pt inter(line r, line s) {
237     if (sgn((r.p - r.q) ^ (s.p - s.q)) == 0)
238         return pt(DINF, DINF);
239     pt rq = r.q - r.p;
240     return rq * get_t(rq, s) + r.p;
241 }
242 // Cuts polygon with line r leaving points
243 // strictly to the left of r
244 vector<pt> cut_polygon(vector<pt> v, line r) {
245     vector<pt> ans;
246     for (int j = 0; j < (int)sz(v); j++) {
247         if (ccw(r.p, r.q, v[j])) ans.push_back(v[j]);
248         if (sz(v) == 1) continue;
249         line s(v[j], v[(j+1)%sz(v)]);
250         pt p = inter(r, s);
251         if (isinseg(p, s)) ans.push_back(p);
252     }
253     ans.erase(unique(all(ans)), ans.end());
254     if (sz(ans) > 1 && ans.back() == ans[0])
255         ans.pop_back();
256     return ans;
257 }
258
259 struct circle {
260     pt c; ld r; // customizable to ll
261     circle(pt x, pt y, pt z) {
262         y = (x+y)/2.0; z = (x+z)/2.0;
263         c = inter(line(y, y + rotate90(x-y)),
264                 line(z, z + rotate90(x-z)));
265         r = dist(c, x);
266     }
267     circle(pt c, ld r) : c(c), r(r) {}
268 };
269 vector<pt> circ_line_inter(const line &l, circle &c) {
270     vector<pt> ans;
271     pt b = l.q - l.p, a = l.p - c.c;
272     ld A = b*b, B = a*b, C = a*a - c.r*c.r;
273     ld D = B*B - A*C;
274     if (sgn(D) < 0) return ans;
275     ans.push_back(c.c+a+b*(-B+sqrt(max((ld)0,D)))/A);
276     if (sgn(D) > 0)
277         ans.push_back(c.c+a+b*(-B-sqrt(max((ld)0,D)))/A);
278     return ans;
279 }
280 vector<pt> circ_inter(circle ca, circle cb) {
281     pt a = ca.c, b = cb.c;
282     ld r = ca.r, R = cb.r, d = dist(a, b);
283     vector<pt> ans;
284     if (sgn(d - (r+R)) > 0 ||
285         sgn(d + min(r, R) - max(r, R)) < 0) return ans;
286     ld x = (d*d - R*R + r*r) / (2.0*d);
287     ld y = sqrt(max((ld)0.0, r*r - x*x));
288     pt v = (b-a)/d;
289     ans.push_back(a + v*x + rotate90(v)*y);
290     if (sgn(y) > 0) ans.push_back(a+v*x-rotate90(v)*y);
291     return ans;
292 }
293 // --- POLYGON & LATTICE (INTEGER SAFE) --- //
294
295 // Centroid of a polygon. Area must be > 0.
296 pt centroid(const vector<pt>& v) {
297     pt c(0, 0); ld A = 0;
298     for (int i = 0; i < (int)sz(v); i++) {
299         ld cross = (v[i] ^ v[(i+1)%sz(v)]);
300         A += cross;
301         c = c + (v[i] + v[(i+1)%sz(v)]) * cross;
302     }
303     return c / (3.0 * A);
304 }
305 // Num of integer lattice points on polygon boundary
306 int boundary_pts(const vector<pt>& v) {
307     int b = 0;
308     for (int i = 0; i < (int)sz(v); i++) {
309         pt p = v[i], q = v[(i+1)%sz(v)];
310         b += gcd(abs((int)(p.x-q.x)), abs((int)(p.y-q.y)));
311     }
312     return b; // Obs: Interior pts I = (2*Area-B+2)/2
313 }
314 // Minkowski sum of two CONVEX polygons - O(N*M)
315 void reorder_pol(vector<pt>& p) {
316     int pos = 0;
317     for(int i = 1; i < (int)sz(p); i++) {
318         if(p[i].y < p[pos].y ||
319            (p[i].y==p[pos].y && p[i].x<p[pos].x)) pos = i;
320     }
321     rotate(p.begin(), p.begin() + pos, p.end());
322 }
323 vector<pt> minkowski(vector<pt> P, vector<pt> Q) {
324     reorder_pol(P); reorder_pol(Q);
325     P.push_back(P[0]); P.push_back(P[1]);
326     Q.push_back(Q[0]); Q.push_back(Q[1]);
327
328     vector<pt> ans;
329     int i = 0, j = 0;
330     while(i < (int)sz(P)-2 || j < (int)sz(Q)-2) {
331         ans.push_back(P[i] + Q[j]);
332         auto cross = (P[i+1] - P[i]) ^ (Q[j+1] - Q[j]);
333         if(cross >= 0 && i < (int)sz(P) - 2) i++;
334         if(cross <= 0 && j < (int)sz(Q) - 2) j++;
335     }
336     return ans;
337 }
338 // O(N) diameter of CONVEX pol by rotating calipers
339 ld diameter(vector<pt> p) {
340     if(sz(p) <= 1) return 0;
341     if(sz(p) == 2) return dist(p[0], p[1]);
342     ld ans = 0;
343     for(int i = 0, j = 1; i < (int)sz(p); i++) {
344         pt nxt = p[(i+1)%sz(p)];
345         while(sarea2(p[i], nxt, p[(j+1)%sz(p)]) >
346              sarea2(p[i], nxt, p[j])) j = (j+1)%sz(p);
347         ans = max({ans, dist(p[i],p[j]), dist(nxt,p[j])});
348     }
349     return ans;
350 }
351 // =====//
352 // --- FLOAT ONLY (REQUIRE 'using T = ld;') --- //
353 // =====//
354
355 struct halfplane {
356     pt p, pq; ld ang;
357     halfplane() {}
358     halfplane(pt a, pt b) : p(a), pq(b - a) {
359         ang = atan2((ld)pq.y, (ld)pq.x);
360     }
361     bool out(pt r) { return sgn(pq ^ (r - p)) < 0; }
362     bool operator<(const halfplane& e) const {
363         if(sgn(ang-e.ang) == 0) return sgn(pq^(e.p-p))<0;
364         return ang < e.ang;
365     }
366     pt inter(const halfplane& s) {
367         pt rq = s.p - p;
368         ld t = (ld)(s.pq ^ rq) / (ld)(pq ^ s.pq);
369         return p + pq * t;
370     }
371 };
372 // Half-Plane Intersection - O(N log N)
373 // Returns convex polygon bounding the intersection
374 vector<pt> hpi(vector<halfplane> H) {
375     sort(all(H));
376     deque<halfplane> q; deque<pt> ans;
377     for(int i = 0; i < (int)sz(H); i++) {
378         while(!ans.empty() && H[i].out(ans.back())) {
379             ans.pop_back(); q.pop_back();
380         }
381         while(!ans.empty() && H[i].out(ans.front())) {
382             ans.pop_front(); q.pop_front();
383         }
384         if(i>0 && sgn(H[i].ang-H[i-1].ang) == 0) continue;
385         q.push_back(H[i]);
386         if(sz(q) > 1)
387             ans.push_back(q.back().inter(q[sz(q)-2]));
388     }
389     while(!ans.empty() && q.front().out(ans.back())) {
390         ans.pop_back(); q.pop_back();
391     }
392     if(sz(q) < 3) return {};
393     ans.push_back(q.back().inter(q.front()));
394     return vector<pt>(all(ans));
395 }
396 // Minimum Enclosing Circle (Welzl's Algorithm)

```

```

397 // O(N) randomized
398 circle welzl(vector<pt> v) {
399     mt19937 rng(1337); // change to chrono
400     shuffle(all(v), rng);
401     circle c(pt(0,0), 0);
402     for (int i = 0; i < sz(v); i++) {
403         if (dist(c.c, v[i]) <= c.r + eps) continue;
404         c = circle(v[i], 0);
405         for (int j = 0; j < i; j++) {
406             if (dist(c.c, v[j]) <= c.r + eps) continue;
407             c = circle((v[i]+v[j])/2.0, dist(v[i],v[j])/2.0);
408             for (int k = 0; k < j; k++) {
409                 if (dist(c.c, v[k]) <= c.r + eps) continue;
410                 c = circle(v[i], v[j], v[k]);
411             }
412         }
413     }
414     return c;

```

```

415 }
416
417 // --- COMPARATORS FOR SWEEP-LINE ALGORITHMS --- //
418
419 // Comparator for lines (assumes p < q)
420 bool operator<(const line& a, const line& b) {
421     pt v1 = a.q - a.p, v2 = b.q - b.p;
422     ld a1 = angle(v1), a2 = angle(v2);
423     if (sgn(a1 - a2) != 0) return a1 < a2;
424     return ccw(a.p, a.q, b.p);
425 }
426
427 bool operator==(const line& a, const line& b) {
428     return !(a < b) && !(b < a);
429 }
430
431 // Sweep-line state comparator
432 // (horizontal sweep, vertical sort)
433 struct cmp_sweepline {
434     bool operator()(const line& a, const line& b) const {

```

```

433     if (a.p== b.p) return ccw(a.p, a.q, b.q);
434     if (sgn(a.p.x - a.q.x) != 0 &&
435         (sgn(b.p.x-b.q.x)==0 || sgn(a.p.x-b.p.x)<0))
436         return ccw(a.p, a.q, b.p);
437     return ccw(a.p, b.q, b.p);
438 }
439 };
440 // Sweep-angle state comparator
441 // (radial sweep, distance sort)
442 struct cmp_sweepangle {
443     pt dir; // Set this globally before comparing
444     bool operator()(const line& a, const line& b) const {
445         return sgn(get_t(dir, a) - get_t(dir, b)) < 0;
446     }
447 };

```